

UNIT- V

ENVIRONMENTAL POLICY, LEGISLATION & EIA

5.1.1. National Environmental policy:

Environmental policy is any [course of] action deliberately taken [or not taken] to manage human activities with a view to prevent, reduce, or mitigate harmful effects on nature and natural resources, and ensuring that man-made changes to the environment do not have harmful effects on humans.

Definition: It is useful to consider that environmental policy comprises two major terms: environment and policy. Environment primarily refers to the ecological dimension (ecosystems), but can also take account of social dimension (quality of life) and an economic dimension (resource management). Policy can be defined as a "course of action or principle adopted or proposed by a government, party, business or individual". Thus, environmental policy focuses on problems arising from human impact on the environment, which retroacts onto human society by having a (negative) impact on human values such as good health or the 'clean and green' environment.

To achieve the goal of clean environment, a two-pronged strategy is required to be made i.e., effective laws to protect the environment coupled with willing co-operation and active participation of the citizens. The need is to have a sound environment policy.

5.1.2. Environmental (protection) Act, 1986 :

1. Objectives of this Act:

- i) Protection and improvement of environment (water, air, land) and,
- ii) Prevention of hazardous to all living creatures (human, plants, animals) and property.
- iii) Maintenance of harmonious relationship between human beings and their environment.

2. Details about the Act:

i) General function of the Central government:

- 1. Protecting and improving the quality of environment
- 2. Preventing and control of pollution
- 3. Laying down standards for quality like emission standards for gaseous and waste water pollution from industries and other organizations
- 4. Restriction of areas for establishing industries

5. Laying down procedures and safe guards for prevention and control of accidents and hazardous operations.
6. Checking by analyzing or monitoring manufacturing process and substances
7. Carrying out and sponsoring research activities
8. Preparation of standards, codes, guides, manuals and other material for education, prevention, control and abatement related to environmental awareness and environmental pollution control.

ii) Provisions in this Act, for the prevention, control and Abatement of environmental pollution:

- a) Under Section 7.1, no person carrying on any industry, operation or process shall be permitted to discharge any environmental pollutant in excess of prescribed standards.
- b) Under Section 8.2, no person shall handle hazardous substances without complying with the prescribed procedural safeguards.
- c) Under Section 9.3, a person is responsible to mitigate environmental pollution and to intimate the fact of any occurrence relating to environmental pollution to the concerned authorities and also assist those authorities in preventing or mitigating environmental pollution. The expenses incurred with respect to remedial measures shall be recoverable from the persons responsible.

iii) Powers of the Central Government:

a) *Power of entry and inspection:* According to Section 10, the Central Govt. officers have the power to enter and inspect any place for the purpose of performing any function entrusted under the legislation.

Any person carrying on an industry, operation or process or handling any hazardous substance shall have to render all assistance to the Central Govt. and its appointed officers failing or obstructing which will be a punishable offence under this Act.

b) *Power to take samples:* Under Section 11, the Central Govt. and its officers have the power to take samples of air, water, soil or substances from the factory or place, for analysis according to the following procedure:

- giving notice to the occupier of the place or his agent.
- collecting samples in the presence of the occupier or his agent.
- placing the sample in a sealed container, signed by the officer and occupier or agent.
- immediately sending one part of the sample to a laboratory for analysis after giving one part to the occupier, and, producing the third part in the court.

In case the occupier or the agent refuses to sign the container, the person collecting the sample may himself do so after reporting the refusal in writing to the government analyst.

c) *Power to establish laboratories:* Under Section 12, the Central Govt. has the power to establish environmental laboratories or recognize any laboratory or institute as an environmental laboratory.

Under Section 13, the Central Govt. has the power to appoint or recognize government analyst for the purpose of analysis of samples of air, water, soil or any other substance.

Under Section 14, the report signed by government analysts may be used as evidence of the facts stated therein in any proceeding under the legislation.

The Central Govt. has the power to close, prohibit or regulate any industry or operation or to stop or regulate the supply of electricity or water or any other service.

d) Penalties for violating the provision in this Act: According to Section 15.5, any person failing to comply with any of the provisions of the Act shall be punishable with imprisonment for a term which may extend up to five years or with fine which may extend to one lakh rupees or both.

According to Section 16, in case of an offence being committed by accompany, the criminal liability is fixed on the company's directors and principal officers also.

Under Section 17, criminal liability is also fixed on the heads of department of government where an offence is committed by the concerned department and the head of the department is unable to prove that the offence was committed without his knowledge or that the exercised all diligence to prevent the commission of such an offence.

For the effective implementation of this Act, the various provisions of the Environment (Protection) Act, 1986 have been amended from time to time by the Government of India.

The chronology of Environment Legislator in India is presented in below Table.

Year	Environmental Legislation
1974	The Water (Prevention and control of pollution) Act
1975	The water (Prevention and control of pollution) Rules
1977	The water (Prevention and control of pollution) Cess Act
1978	The water (Prevention and control of pollution) Cess Rules
1981	The Air (Prevention and control of pollution) Act
1982/1983	The Air (Prevention and control of pollution) Rules
1986	The environment (Protection) Act
1989	The Hazardous waste (Management and Handling) Rules
1989/1994	Manufacture, storage and Import of Hazardous chemical Rules/Amendment Rules

It may be noted that as per the Constitution of India the legislative power to make laws in relation to water vests with the States. But as per the provisions of the Constitution, the States passed resolutions authorizing the Parliament to pass the Water (Prevention and Control of Pollution) Act, 1974. The Water Act deals with the prevention and control of water pollution, and the Air Act is concerned with the prevention and control of air pollution. The Environment Act was enacted as a comprehensive legislation for the protection and improvement of the

environment. In 1991, a new legislation, namely, the Public Liability Insurance Act was enacted, for the purpose of providing immediate relief to the persons affected by accident occurring while handling any hazardous substances.

5.1.3. The Air (Prevention and Control of Pollution) Act, 1981:

i) Objectives of the Act:

This Act has been passed to provide for the following:

- a) Prevention, control and abatement of air pollution
- b) Maintaining the quality of air
- c) Establishment of boards for the prevention and control of air pollution.

ii) Definition of pollution under this Act:

‘Air-pollution’ means the presence in the atmosphere of any air pollutant.

‘Air-pollutant’ means any solid, liquid or gaseous substances (including noise) present in the atmosphere in such concentrations as may be or tend to be injurious to human beings or other living creatures or property or environment.

‘Control equipment’ means any apparatus, device, equipment or system to control the quality and manner of emission of any air pollutant and includes any device used for securing the efficient operation of any industrial plant.

iii) Powers and Functions of Boards:

1. Constitution and Functions of Central Boards: To have an integrated approach for tackling the problems related to pollution, this Act provides that the Boards for the prevention and control of water pollution, constituted under Section 3 and 4 of the Water (Prevention and Control of Pollution) Act, 1974, shall also act as Air pollution Control Boards under the Act as:

- a) The Central Board for the Prevention and Control of Air Pollution under Section – 3
- b) The State Board for the Prevention and Control of Air Pollution under Section- 4

The functions of the various boards for control of air pollution are also identical to the functions of the Boards for Control of water Pollution except that they aim to control air pollution whereas the Boards Under Water Pollution Act aim to control water pollution.

2. Powers of the Board:

a) Power to declare Air pollution control area: Under Section- 19.1, the State Government, after consultation with State Boards may:

- i) declare any area or areas as Air Pollution Control Areas by notification in the official gazette.
- ii) prohibit use of any fuel or appliance causing or likely to cause air pollution in an Air Pollution Control Area.

iii) prohibit burning of any material causing or likely to cause pollution in an Air Pollution Control Area.

b) Power to establish standards for emission of air pollution from automobiles: Under Section 20.2 the Boards have the power to ensure that the standards for emission of air pollutants, from automobiles laid down by the State Boards under clause (vii) of sub-section (1) of Section-17 are complied with.

c) Power to restrict use of certain industrial plants: Under Section 21.3 of this Act, no person shall, without the previous consent of the State Board, establish or operate any industrial plant in an Air Pollution Control Area, and, Under Section 22.4 of this Act, no person operating any industrial plant in any Air Pollution Control Area shall discharge or cause or permit to be discharged, the emission of any air pollutant in excess of the standards laid down by the State Board under clause (vii) of sub-section (1) of Section 17.

d) Powers of entry and inspection: Under Section 24.5 of this Act, any person empowered by a State Board shall have the right to enter any place for

i) performing any of the functions of the State Board entrusted to him.

ii) examining and testing any control equipment, industrial plant, record, register, documents etc.

iii) seizing any such control equipment if he has reasons to believe that it may furnish evidence of the commission of an offence punishable under this Act.

e) Power to take samples: Under Section 26 of this Act, any officer empowered by the State Board shall have the power to take, for the purpose of analysis, samples of air or emission from any chimney, flue or duct etc. and send the same for analysis to the laboratory established or recognized by the State Board.

3. Penalties for violation of the provisions under the Act: Except for the fact that there is no provision for publication of names of offenders under this Act, the penalties for defaults or violations or the provisions under the Act are the same as under the water Act.

The main drawback of these Acts lies in giving the defaulter 60 days' notice before taking him to court as by the time the case is filed in court, the offender may destroy the evidence, in which case it becomes difficult to prove the offence. Another drawback is that no consent or permission is required to be taken from the Board for establishing industry outside the Air Pollution Control Area even though its emission may be reaching the Air Pollution Control Area also.

Yet another drawback lays in the fact that discharge of any air pollutant into the atmosphere by a ship or aircraft is excluded from the purview of the Act.

5.1.4. The Water (Prevention and Control of Pollution) Act-1974:

i) Objectives of this Act:

- a) Prevention and control of water pollution
- b) Maintaining or restoring the wholesomeness of water
- c) Establishment of Boards of the Prevention and Control of water pollution.

ii) Definitions of 'Pollution' Under the Act:

“Pollution” has been elaborately defined as the contamination of water, or the alteration of the physical, chemical or biological properties of water, or the discharge of any sewage or trade effluent (whether directly or indirectly) which is likely to render such water harmful or injurious to a) public health or safety b) domestic, commercial, industrial, agricultural or other uses c) life and health of plants, animals or aquatic organisms.

iii) Powers and Functions of Boards:

1. Constitution of Central Board: According to Section-3, the Central Board shall consist of the following members:

- a) A full time Chairman (to be maintained by the Central Government) having knowledge or practical experience in matters related to environmental protection or having knowledge and experience in administration of institutions dealing with matters aforesaid.
- b) Not more than five officials nominated by the Central Government.
- c) Not more than five persons nominated by the Central government from amongst the members of State Boards.
- d) Not more than three non-officials nominated by the government to represent interests of agriculture, fishery, agriculture-trade etc.
- e) Two persons nominated by the government to represent the companies or corporations owned by the Central Government.
- f) One full time Member-Secretary (to be appointed by the Central Government) having knowledge and experience of scientific engineering or management aspects of pollution control.

2. Constitution of State Boards: Under Section-4, of the State Pollution Control Board may be constituted having the same constitution as the Central Board.

3. Constitution of Committees: A board may constitute as many committees as necessary. The members of a committee shall be paid such fees and allowances for attending to any other work of the Board.

4. Constitution of Joint Boards: according to Section – 14 of the Act, under agreement between two or more contiguous States, Joint Boards may be constituted for those states, by the Central or State Governments.

5. Terms and Service conditions of the members of the Board:

- a) Terms and service conditions of the Member Secretary and Chairman shall be as prescribed by the Government.
- b) Rest of the members shall hold office for a term of three years.
- c) A member shall be eligible for re-nomination.
- d) Central or State Government may remove a member of the Central or State Board at any time by giving him reasonable notice and opportunity.
- e) The Chairman may resign by addressing his resignation to the government and a member may resign by addressing his resignation to the Chairman.
- f) In the case of insolvency, unsound mind, conviction for the offence under this Act, conviction for the offence involving moral turpitude, inability to attend three consecutive

meetings abusing position as member of the Board, Partnership with anybody dealing with sewage or trade effluent etc. are some conditions for disqualification of the member. Seat of the disqualified member shall fall vacant and a person nominated to fill such a vacancy shall hold office for the remaining term.

6. Functions of the Central Board: According to Section 16-A, the following are the functions of the Central Board:

- a) to promote cleanliness of streams and wells in different areas of the state.
- b) to advise the Central Government on matters concerning the prevention and control of water pollution.
- c) to co-ordinate the actions of the State Board and resolve disputes among them.
- d) to provide technical assistance and guidance to the State Boards to carry out research in prevention and control of water pollution problems.
- e) to organize training of persons engaged in pollution control.
- f) to organize comprehensive program for pollution control through mass media.
- g) to lay down standards for streams or wells.
- h) to prepare manuals, codes or guides for treatment and disposal of sewage and trade effluents.
- i) to establish or recognize laboratories for analysis of water samples from any streams, well or trade effluents.

7. Functions of the State Boards: Under Section 7-B, the following are the functions of a State Board:

- a) Planning a comprehensive program for prevention, control and abatement of pollution of streams and wells.
- b) Advising the State Government regarding water pollution control or location of industries.
- c) Conduction and encouraging investigations and research relating to different aspects of water pollution.
- d) To collaborate with the Central Board for training personnel for handling water pollution programs and organizing related mass education programs.
- e) Inspecting trade effluents and waste treatment plants.
- f) Prescribing effluent standards for the sewage and trade effluents.
- g) Evolving economical and reliable methods of disposal, treatment and reuse of waste water (in agriculture).
- h) Laying down the standards of treatment of sewage and trade effluents to be discharged into any stream.
- i) Making, varying or revoking any order for preservation of control of discharge of waste into streams and wells or construction of systems for disposal of effluents.
- j) Establishing or recognizing laboratories for analysis of samples.
- k) Performing such functions as may be entrusted by Central Board or State Governments.

8. Powers of the State Government:

- a) Power to obtain information: According to Section 20.2, the State Board may make surveys, take measurements or obtain information for purpose of performing functions under this Act. Failure to comply with any directions under the Sections is a punishable offence under substance (1) of Section 41.

b) Power to take samples: Under Section 21 (1)A, the State Government has the power to take samples of water of any stream or well or any effluent being discharges into such a stream or well, for analysis. Under Section 22.4, the State Board further has the power to obtain a report of the result of the analysis by a recognized laboratory.

c) Power of entry and inspection: according to Section 23.5, the State Board is empowered by the State Government, with the right to enter any place for the purpose of performing any of the functions entrusted to it.

d) Power of prohibition on disposal of polluting matter into a stream or well: Under Section 24.6, i) no person shall knowingly allow entry of any poisonous, noxious or polluting matter directly or indirectly into any streams, well or sewer or on land.

ii) no person shall knowingly allow entry of any matter into ant stream, which may impede the proper flow of water resulting in substantial aggravation of pollution.

According to Section 25.7, without the proper consent of the State Board, i) no person shall establish any industry, operation or process or any treatment disposal system, which is likely to discharge any sewage or effluent into stream or well or on land.

ii) no person shall use any new outlet for discharge of sewage.

iii) no person shall begin to make any new discharge of sewage.

9. Penalties for violation of the provisions of the Act:

a) In case of failure to give information by a person discharging effluents into stream or well or regarding construction or establishment of a disposal system the penalty is imprisonment up to 3 months or fine up to Rs. 10,000/- or both.

b) In case of destroying or damaging the property of the Board, obstructing the performance of the Board's functions, failure to furnish information about accidents under Section 31, giving wrong information or making false statements to get Board's consent, the penalty is imprisonment up to 3 months or fine up to Rs. 10,000/- or both.

c) In case of violation of order prohibiting discharge of any polluting matter into stream, well or land, or violation of control order restraining pollution of water or streams or wells or violation of Board order of closure of industry or stoppage of water or electricity supply etc., the penalty is imprisonment for one and a half years to six years and fine. If it continues there is an additional fine of Rs. 5000/- per day and if it continues beyond one year then the penalty is imprisonment of two to seven years on conviction.

d) In case of permitting material into any stream, well or land the penalty is imprisonment for one and a half years to six years or fine or both.

5.1.5. Forest (Conservation) Act, 1980 with amendments made in 1988 :

1. Objectives:

This Act has been passed to provide for the following objectives:

- a. Protection and conservation of forests.
- b. to ensure judicious use of forest products.

2. Details of the Act:

- a. Notification of reserve forest:

According to section-3 of this Act a forest land or waste land which is property of the government and whole or part of the forest produce may be constituted as a Reserve forest by the State Government after issuing notification in the official gazette under section 4 of this Act.

- a) Declaring that it has been decided to constitute any land as a reserve forest.
- b) Specifying the situation and limits of such a land.
- c) Appointing an officer not holding any other forest office except that of Forest Settlement Officer.

According to Section 6 of this Act, the proclamation shall be such so as to:

- a) Specify the situation and limits of the proposed forest.
- b) Explain the consequences which will ensure the reservation of such forest.
- c) Fix a period of not less than 3 months from the date of such proclamation for claiming any right over any land or over forest produce to present to the Forest Settlement Officer a written notice specifying the nature of such right and amount and particulars of the compensation claimed.

3. Powers of the Forest Officer:

i. Power to Stop ways and water courses in reserved forest: Under Sec. 5 of this Act, with previous sanction of the State Government of the Forest Officer may stop any public or private way or water course in a reserved forest.

ii. Power to issue notification declaring protected forests and making rules for such forest: Under Sec.29 of this act, the state government by notification in the official gazette may declare the provisions of this Act applicable to any forest-land or waste land not included in a reserved forest but which is property of the government or to the whole or part of forest produce.

Under sec. 32 of the Act, the State Govt. may make rules for protection of forest.

iii. Power to issues notification reserving trees: Under Sec. 30 of this act, by notification in the official gazette, the State Govt. may a) declare any tree or trees in a protected forest as reserved. b) declare any portion of such forest specified in the notification as closed for not more than thirty years. c) prohibit the quarrying of stone or the burning of lime or charcoal or collection or removal of any forest produce.

iv. Power to declare prohibition in forests: Under sec.35 by notification in the official gazette, the state govt. may prohibit the following in any forest or waste-land: a. the breaking – up or clearing of land for cultivation. B) the pasturing of cattle c) clearing of the vegetation etc.

v. Power to assume management of forests: In case of any prohibition under sec. 35 is violated, the State Govt. under sec.36, may after giving notice in writing to the owner of such forest land and after considering his objection, assure the management of forests by placing the same under the control of a forest officer.

vi. Power to form village forests: Under sec.28 the state govt. may assign to any village community the rights of govt. over any land that has been constituted as a reserve forest and may cancel such assignment. Such forest shall be called as village forest.

4. Penalties for violation of the provisions in this act:

Under sec. 33 of this Act, imprisonment for a term of six months or more or fine of five hundred rupees or more or both, is the penalty for any person who commits any of the following offences:

- a. fells, girdles, lops, taps or burns any tree reserved under sec. 30 or damages any such trees.

- b. violates the prohibition under sec. 30, by clearing for cultivation etc. any land in any protected forest.
- c. violates the prohibition under sec. 30, by quarrying any stone, burning any lime or charcoal or collecting or removing any forest produce.
- d. sets fire to such forest or kindles fire without taking all reasonable precaution to prevent its spreading to any tree reserved under sec. 30, whether standing, fallen or felled, to any closed portion of such forest.
- e. permits cattle to damage any reserved tree.
- f. violates any of the rules under sec. 32.
- g. A person who kills or catches elephants in any area in which the Elephant's Preservation Act, 1879, is not in force, is punishable with imprisonment for a term which may extend to six months or with fine which may extend to five thousand rupees or both, in addition to compensation for damage done to the forest.
- h. Under sec. 55, all timber or forest produce which is not the property of the government and in respect of which a forest offense has been committed and all tools, carts, cattle etc. used for committing the offense are liable to confiscation in addition to punishment prescribed for such an offense.

The existing environmental protection laws have failed to prove effective in controlling the environmental pollution to the desired level. The reason for this is many, namely,

- a. There is gross violation of rules, regulations and provisions in the Acts.
- b. The laws are on specific types of pollution and are outdated.
- c. Some major areas of environmental hazards are not covered by them
- d. The laws lack statements of explicit policy objectives.
- e. The laws lack adequate provisions for implementing them.
- f. There is no set procedure and time limit for reviewing the performance of the laws.

There is need for an authority to:

- a. Co-ordinate the activities of other agencies
- b. Frequently review these legislations
- c. Study, plan and implement long term requirement of environmental safety keeping in view the extent of environmental pollution and its hazards.
- d. Ensure speedy and efficient handling of emergency situations threatening the environmental degradation.

In view of all these shortcomings and requirements, the parliament passed the Environment (Protection) Act in 1986 as a general legislation.

5.1.6. Wildlife Protection Act, 1972 (amended in 2002)

1. Definition of Wildlife:

The Section 2(37) of the Act defines wildlife as wildlife includes any animal, bee's butterflies, Crustacean, fish and moths; and aquatic or land vegetation which forms part of any habitat. So the meaning of the wildlife in this Act is very wide and inclusive of all kinds of flora and fauna.

2. Authorities Constituted Under Wildlife (Protection) Act:

As per the Sec. 3 of the Act, the Central Government may appoint a Director of Wildlife Preservation, Assistant Directors of Wildlife Preservation and such other officers and employees as may be necessary.

As per the Sec. 4, the State Government may, for the purpose of this Act, appoint Chief Wildlife Warden, Wildlife Warden, Honorary Wildlife Wardens and other officers and employees as may be necessary.

As per Sec. 6, the State Governments and the Administrators in Union Territories shall constitute a Wildlife Advisory Board.

3. The Wildlife Advisory Board (Sec. 6):

The Sec. 6 of this Act enforces and enables the state governments and the administrators of the Union Territories to constitute a Wildlife Advisory Board in each states and union territories.

4. The members of the Wildlife Advisory Board:

It shall consist of the Minister in charge of Forests in the State or Union territory as the Chairman. If there is no such minister, then the Chief Secretary will be the Chairman of the Board. The other members are, two members of the State Legislature or Legislature of Union Territory; Secretary to the state government or the government of the union territory, in charge of forests; the Forest Officer in charge of the State Forest Department; an officer nominated by the Director of Wildlife Preservation; Chief wildlife warden; officers of the state government not exceeding 5; and such other persons, not exceeding 10, who in the opinion of the state government, are interested in the protection of wildlife, including the representatives of tribal not exceeding 3.

5. Duties of Wildlife Advisory Board (Sec. 8):

The Wildlife Advisory Board mainly constituted to advise the state government in the following matters.

- a) In the selection of areas to be declared as Sanctuaries, National Parks and Closed areas and the administration thereof;
- b) In formulation of the policy for protection and conservation of wildlife and specified plants;
- c) In any matter relating to the amendment of any schedule;
- d) In relation to the measure to be taken for harmonizing the needs of the tribal and other dwellers of the forests with the protection and conservation of wildlife;
- e) In any other matter connected with the protection of wildlife which may be referred to it by the state government.

6. Hunting of Wild Animals (Sec. 9):

Sec. 9 of the Act prohibits hunting of any wild animal specified in Schedules 1, 2, 3, and 4. Any person who hunts any wild animal shall be punishable with imprisonment for a term which may extend to 3 years or with fine which may extend to Rs. 25000/- or with both. However if any person commits the offence in the sanctuary or national park, with respect any animal

specified in Schedule 1, he shall be punishable with imprisonment which shall not be less than 1 year but may extend to 6 years and also with fine which shall not be less than 5000/-.

7. Hunting of Wild animals to be permitted in certain cases:

The Chief Wildlife Warden may permit hunting of wild animals in certain situations. They are;

(a) The Chief Wildlife Warden may, if he is satisfied that any wild animal specified in Schedule 1 has become dangerous to human life or is so disabled or diseased as to be beyond recovery, by order in writing and stating the reasons therefore, permit any person to hunt such animal or cause animal to be hunted;

(b) The Chief Wildlife Warden or the authorized officer may, if he is satisfied that any wild animal specified in Schedule. II or III or IV has become dangerous to human life or to property (including standing crops on any land) or is so disabled or diseased as to be beyond recovery, by order in writing and stating the reasons therefore, permit any person to hunt such animal or cause such animal to be hunted.

(c) The killing or wounding in good faith of any wild animal in defense of oneself or of any other person shall not be an offence; Provided that nothing in this sub-section shall exonerate any person who, when such defense becomes necessary, was committing any act in contravention of any provision of this Act or any rule or order made there under.

(d) Any wild animal killed or wounded in defense of any person shall be Government property.

8. Grant of permission for hunting for special purposes:

The Chief Wildlife Warden, permit, by an order in writing stating the reasons therefore, to any person, on payment of such fee as may be prescribed, which shall entitle the holder of such permit to hunt, subject to such conditions as may be specified therein, any wild animal specified in such permit, for the purpose of,

(a) Education;

(b) Scientific research;

(c) Scientific management; means and includes

(i) Translocation of any wild animal to an alternative suitable habitat; or

(ii) Population management of wildlife, without killing or poisoning or destroying any wild animals.

(d) Collection of specimens

(i) for recognized zoos subject to the permission under section 38-1 or

(ii) for museums and similar institutions;

(e) Derivation, collection or preparation of snake-venom for the manufacture of life saving drugs.

9. Protection of Specified plants:

Sec. 17A of the Act prohibits picking, uprooting, etc., of specified plants.

No person shall:

(a) will fully pick, uproot, damage destroy, acquire or collect any specified plant from any forest land and area specified, by notification, by the Central Government,

- (b) possess, sell, offer for sale, or transfer by way of gift or otherwise, or transport any specified plant, whether alive or dead, or part or derivative thereof:

10. Sanctuaries, National parks, Zoo parks:

Section 18 provides that the State Government may, by notification, declare its intention to constitute any area other than area comprised with any reserve forest or the territorial waters as a sanctuary, national park if it considers that such area is of adequate ecological, faunal, floral, geomorphological, natural or zoological significance, for the purpose of protecting, propagating or developing wildlife or its environment.

11. Punishments:

Provided that where the offence committed is in relation to any animal specified in Scheduled I or Part II of Schedule. II, or meat of any such animal, animal article, trophy, or uncured trophy derived from such animal or where offence [relates to hunting in, or, altering the boundaries of] a sanctuary or a National Park, such offence shall be punishable with imprisonment for a term which shall not be less than [one year] but may extend to six years and also with fine which shall not be less than five thousand rupees. Provided further that in the case of a second or subsequent offence of the nature mentioned in this sub-section, the term of imprisonment may extend to six years and shall not be less than two years and the amount of fine shall not be less than ten thousand rupees.

Any person who contravenes any provisions of Chapter VA, [Prohibition of Trade or Commerce in Trophies, Animal Articles, etc. derived from Certain Animals.] shall be punishable with imprisonment for a term which shall not be less than one year but which may extend to seven years and also with fine which shall not be less than five thousand rupees.

Any person who contravenes the provisions of Section 38J [tease, molest, injure or feed any animal or cause disturbance to the animals by noise or otherwise, or litter the grounds in a zoo] shall be punishable with imprisonment for a term which may extend to six months or with fine which may extend to two thousand rupees, or with both. Provided that in case of second or subsequent offence the term of imprisonment may extend to one year or the fine may extend to five thousand rupees as per section 52.

5.1.7. The Municipal Solid Wastes (Management and Handling) Rules, 2000.

Central Government notified in exercise of the Powers conferred by Section 3, 6 and 25 of the Environment (Protection) Act, 1986 with the objective of regulating the management and handling of the Municipal Solid Wastes.

1. Applicability:

Apply to all municipal authorities responsible for collection, segregation, storage, transportation processing and disposal of municipal solid wastes.

2. Responsibility of municipal authority:

1. Every municipal authority shall, within the territorial area of the municipality, be responsible for the implementation of the provisions of these rules, and for any infrastructure development for collection, storage, segregation, transportation, processing and disposal of municipal solid wastes.
2. the municipal authority or an operator of a facility shall make an application in Form-I, for grant of authorization for setting up waste processing and disposal facility including landfills from the State Board or the Committee in order to comply with the implementation program laid down in Schedule-I (implementation schedule).
3. the municipal authority shall comply with these rules as per the implementation schedule laid down in Schedule-I.
4. The municipal authority shall furnish its annual report in Form-II.
 - (a) to the Secretary-in charge of the Department of Urban Development of the concerned State or as the case may be of the Union Territory, in case of a metropolitan city; or
 - (b) to the District Magistrate or the Deputy Commissioner concerned in case of all other town and cities, with a copy to the State Board or the Committee on or before the 30th day of June every year.

3. Management of municipal solid waste:

- (1) Any municipal solid waste generated in a city or a town, shall be managed and handled in accordance with the compliance criteria and the procedure laid down in Schedule-II
- (2) The waste processing and disposal facilities to be set up by the municipal authority on their own or through an operator of a facility shall meet the specifications and standards as specified in Schedule-III(specifications for land fill sites) and Schedule-IV (Standards for composing treated lechates and incineration).

Schedule II:

Collection of Municipal Solid Wastes:

1. Littering of municipal solid waste shall be prohibited in cities, town and in urban areas notified by the State Governments. To prohibit littering and facilitate compliance, the following steps shall be taken by the Municipal Authority, namely:-
 - (i) Organizing house-to-house collection of municipal solid wastes through any of the methods, like community bin collection (central bin), house-to-house collection, collection on regular pre-informed timings and scheduling by using bell ringing of musical vehicle (without exceeding permissible noise levels);
 - (ii) Devising collection of waste from slums and squatter areas or localities including hotels, restaurants, office complexes and commercial areas;

- (iii) Wastes from slaughter houses, meat and fish markets, fruits and vegetable markets, which are biodegradable in nature shall be managed to make use of such wastes;
 - (iv) Bio-medical wastes and industrial wastes shall not be mixed with municipal solid wastes and such wastes shall follow the rules separately specified for the purpose;
 - (v) Collected waste from residential and other areas shall be transferred to community bin by hand driven containerized carts or other small vehicles;
 - (vi) Horticultural and construction or demolition wastes or debris shall be separately collected and disposed off following proper norms. Similarly, wastes generated at dairies shall be regulated in accordance with the State laws;
 - (vii) Waste (garbage, dry leaves) shall not be burnt;
 - (viii) Stray animals shall not be allowed to move around waste storage facilities or at any other place in the city or town and shall be managed in accordance with the State laws.
2. The Municipal authority shall notify waste collection schedule and the likely method to be adopted for public benefit in a city or town
 3. It shall be the responsibility of generator of waste to avoid littering and ensure delivery of wastes in accordance with the collection and segregation system to be notified by the municipal authority as per Para 1(2) of the Schedule.

Segregation of Municipal Solid Wastes:

In order to encourage the citizens, municipal authority shall organize awareness programs for segregation of wastes and shall promote recycling or reuse of segregated materials. The municipal authority shall undertake phased program to ensure community participation in waste segregation. For this purpose, regular meetings at quarterly intervals shall be arranged by the municipal authorities with representatives of local resident welfare associations and non-governmental organizations.

Storage of Municipal Solid Wastes:

Municipal authorities shall establish and maintain storage facilities in such a manner as they do not create unhygienic and insanitary conditions around it. following criteria shall be taken into account while establishing and maintaining storage facilities, namely:-

- (i) Storage facilities shall be created and established by taking into account quantities of waste generation in a given area and the population densities. A storage facility shall be so place that it is accessible to users.
- (ii) Storage facilities to be set up by municipal authorities or any other agency shall be so designated that wastes stored are not exposed to open atmosphere and shall be aesthetically acceptable and user-friendly;
- (iii) Storage facilities or 'bins' shall have 'easy to operate' design for handling, transfer and transportation of waste bins for storage of bio-degradable wastes shall be painted green, those for storage of recyclable wastes shall be painted white and those for storage of other wastes shall be painted black;
- (iv) Manual handling of waste shall be prohibited. If unavoidable due to constraints, manual handling shall be carried out under proper precaution with due care for safety of workers.

Transportation of Municipal Solid Wastes:

Vehicles used for transportation of wastes shall be covered. Waste should not be visible to public, nor exposed to open environment prevention their scattering. the following criteria shall be met, namely:-

- (i) The storage facilities set up by municipal authorities shall be daily attended for clearing of wastes. The bins or containers wherever placed shall be cleaned before they start overflowing;
- (ii) Transportation vehicles shall be so designed that multiple handling of wastes, prior to final disposal, is avoided.

Processing of Municipal Solid Wastes:

Municipal authorities shall adopt suitable technology or combination of such technologies to make use of wastes so as to minimize burden on land fill. Following criteria shall be adopted, namely:-

- (i) The biodegradable wastes shall be processed by composting vermin-composting, anaerobic digestion or any other appropriate biological processing for stabilization of wastes. It shall be ensured that compost or any other end product shall comply with standards as specified in Schedule-IV;
- (ii) Mixed waste containing recoverable resources shall follow the route of recycling. Incineration with or without energy recovery including pelletization can also be used for processing wastes in specific cases. Municipal authority or the operator of a facility wishing to use other state-of-the-art technologies shall approach the Central Pollution Control Board to get the standards laid down before applying for grant of authorization.

Disposal of Municipal Solid Wastes:

Land filling shall be restricted to non-biodegradable, inert waste and other waste that are not suitable either for recycling or for biological processing. Land filling shall also be carried out for residues of waste processing facilities as well as pre-processing rejects from waste processing facilities. Land filling of mixed waste shall be avoided unless the same is found unsuitable for waste processing. Under unavoidable circumstances or till installation or alternate facilities, land filling shall be done following proper norms. Landfill sites shall meet the specifications as given in Schedule-II.

Schedule-III

Specification for land filling

1. Waste subjected to land filling shall be compacted in thin layers using landfill compactors to achieve high density of the wastes. In high rainfall areas where heavy compactors cannot be used, alternative measures shall be adopted.
2. Wastes shall be covered immediately or at the end of each working day with minimum 10cm of soil, inert debris or construction material till such time waste processing facilities for composting or recycling or energy recovery are set up as per [Schedule-I](#).
3. Prior to the commencement of monsoon season, an intermediate cover of 40-65 cm thickness of soil shall be placed on the land-fill with proper compaction and grading to prevent infiltration during monsoon. Proper drainage beams shall be constructed to divert run-off away from the active cell of the landfill.
4. After completion of landfill, a final cover shall be designed to minimize infiltration and erosion. The final cover shall meet the following specifications, namely:-
 - (a) The final cover shall have a barrier soil layer comprising of 60 cms of clay or amended soil with permeability coefficient less than 1×10^{-7} cm/sec.
 - (b) On top of the barrier soil layer, there shall be a drainage layer of 15 cm.

-
- (c) On top of the drainage layer, there shall be a vegetative layer of 45 cm to support natural plant growth and to minimize erosion.

5.1.8. Bio Medical Waste(Management & Handling) Rules, 1998:

With a view to control the indiscriminate disposal of hospital waste/bio medical waste, the Ministry of Environment & Forest, Govt. of India has issued a notification on Bio Medical Waste Management under the Environment (Protection) Act. Govt. of NCT of Delhi in its notification dated 6th July, 1999 has authorized Delhi Pollution Control Committee (DPCC) for the purpose of granting authorization for collection, reception, storage, treatment and disposal of bio medical waste to implement the Bio Medical Waste Management Rules, 1998. Govt. of NCT of Delhi has also constituted advisory committee, appellate authority in exercise of powers conferred under Bio Medical rules. Some of the salient features of these rules are:

1. Rules are applicable to: These Rules will apply to hospitals, Nursing Homes, Veterinary Hospitals, Animal Houses, Pathological labs & blood banks, generating hospital wastes. (except such occupier of clinics, dispensaries, pathological labs, blood banks providing treatment/service to less than 1000 (one thousand) patients per month).

2. Duty: It shall be the duty of the every occupier of an institution generating bio medical waste which includes a hospital, nursing home, clinic, dispensary, Veterinary institution animal house, pathological laboratory, blood bank by whatever name called to take all steps to ensure-that such waste is handled without any adverse effect to the human health and the environment.

3. Management of Biomedical waste: Every occupier generating the bio-medical waste need to install an appropriate facility in the premises or set up a common facility to ensure requisite treatment of waste by 30.6.2000 in accordance with Schedule-I and in compliance with standards prescribed with Schedule-V (Standards for incinerators, Standards for waste autoclaving, Standards for liquid waste, Standards for microwaving, Standards for deep burial).

The biomedical waste needed to be segregated into container/bags at the point of generation in accordance with Schedule-II, prior to its storage, transportation, treatment and disposal. The container shall be labeled according to Schedule-III.

4. Penalty: The defaulting hospitals/nursing homes etc. are liable to be penalized as per the provisions of Environment (Protection) Act, 1986 and other pollution control Acts.

5. Appeal: Any person aggrieved by an order made by the DPCC under these rules may within thirty days from date on which the order is communicated to him, prefer an appeal to the Financial Commissioner, Govt. of NCT of Delhi who is appointed as Appellate Authority under the rules.

Schedule-I

Option	Waste category	Treatment and disposal

Category No. 1	Human Anatomical Waste (human tissues, organs, body parts)	incineration @/deep burial*
Category No. 2	Animal Waste (animal tissues, organs, body parts carcasses, bleeding parts, fluid, blood and experimental animals used in research, waste generated by veterinary hospitals, colleges, discharge from hospitals, animal houses)	incineration@/deep burial*
Category No. 3	Microbiology & Biotechnology Waste (Wastes from laboratory cultures, stocks or micro-organisms live or vaccines, human and animal cell culture used in research and infectious agents from research and industrial laboratories, wastes from production of biological, toxins, dishes and devices used for transfer of cultures)	local autoclaving/micro-waving/incineration@
Category No. 4	Waste Sharps (needles, syringes, scalpels, blade, glass, etc. that may cause puncture and cuts. This includes both	disinfection (chemical treatment @@@/auto clawing / microwaving and mutilation / shredding##

	used and unused sharps)	
Category No. 5	Discarded Medicines and Cytotoxic drugs (Waste comprising of outdated, contaminated and discarded medicines)	incineration@/destruction and drugs disposal in secured landfills
Category No. 6	Soiled Waste (items contaminated with blood, and body fluids including cotton, dressings, soiled plaster casts, lines, bedding, other material contaminated with blood)	Incineration @ autoclaving/microwaving
Category No. 7	Solid Waste (Waste generated from disposal items other than the sharps such a tubing's, catheters, intravenous sets etc.)	disinfection by chemical treatment@@ autoclaving/microwaving and mutilation/shredding##
Category No. 8	Liquid Waste (Waste generated from laboratory and washing, cleaning, housekeeping and disinfecting activities)	disinfection by chemical treatment@@ and discharge into drains
Category No. 9	Incineration Ash Ash from incineration of any bio-medical waste)	disposal in municipal landfill
Category No. 10	Chemical Waste (Chemicals used in production of biological, chemicals used in production of biological, chemicals used in disinfection, as insecticides, etc.)	chemical treatment@@ and discharge into drains for liquids and secured landfill for solids.

Note :

- @ There will be no chemical pretreatment before incineration. Chlorinated plastics shall not be incinerated.
- * Deep burial shall be an option available only in towns with population less than five lakhs and in rural areas.
- @@ Chemicals treatment using at least 1% hypochlorite solution or any other equivalent chemical reagent. It musts be ensured that chemical treatment ensures disinfection.
- ## Mutilation/shredding must be such so as to prevent unauthorized reuse.

Schedule-II

COLOUR CODING AND THE TYPE OF CONTAINER FOR DISPOSAL OF BIO MEDICAL WASTES

COLOUR CODING	Type of container	Waste category	Treatment options as per schedule
			Incineration/deep burial
Yellow	Plastic bag	Category 1, 2 and Category 3, 6	
Red	Disinfected container/Plastic bag	Category 3, 6, 7	Autoclaving/Microwaving/ Chemical Treatment
Blue/White translucent	Plastic bag/puncture proof container	Category 4, 7	Autoclaving/Microwaving Chemical Treatment and destruction/shredding
Black	Plastic bag	Category 5, 9 and Category 10 (solid)	Disposal in secured landfill

Note :

- Color coding of waste categories with multiple treatment options as defined in Schedule-I shall be selected depending on treatment option chosen, which shall be as specified in Schedule-I.
- Waste collection bags for waste types needing incineration shall not be made of chlorinated plastics.
- Categories 8 and 10 (liquid) do not require containers/bags.
- Category 3 if disinfected locally need not be put in containers/bags.

Schedule-III

LABEL FOR BIO-MEDICAL WASTE CONTAINERS/BAGS

BIOHAZARD SYMBOL



CYTOTOXIC HAZARD SYMBOL



HANDLE WITH CARE

Note : Labels shall be non-washable and prominently visible

5.1.9. The Hazardous Wastes (Management and Handling) Rules, 1989 (as amended, May, 2003):

1. Responsibility of the occupier and operator of facility for handling of the wastes:

a) The occupier and the operator of a facility shall be responsible for proper collection, reception, treatment, storage and disposal of hazardous wastes listed in Schedule 1, 2 &3.

(b) The occupier or any other person acting on his behalf who intends to get his hazardous waste treated by the operator of a facility under sub-rule(1), shall give, to the operator of a facility, such information as may be specified by the State Pollution Control Board.

(c) It shall be the responsibility of the occupier and the operator of a facility, to take all steps to ensure that the wastes listed in schedules-1, 2 and 3 are properly handled, and disposed of without any adverse effects to the environment.

2. Duties of the occupier and operator of a facility:

It shall be the duty of the occupier and the operator of a facility to take adequate steps while handling hazardous waste to-

(i) Contain contaminants and prevent accidents and limit their consequences on human and the environment; and

(ii) provide persons working on the site with information, training and equipment necessary to ensure their safety.

3. Grant of authorization for handling hazardous waste:

(i) Hazardous wastes shall be collected, treated, stored and disposed of only in such facilities as may be authorized for this purpose

(ii) Every occupier handling, or a recycler recycling, hazardous wastes shall make an application in Form 1 to the Member Secretary, State Pollution Control Board or Committee.

(iii) Any person who intends to be an operator of a facility for the collection, reception, treatment, transport, storage and disposal of hazardous wastes, shall make an application in Form 1 to the Member Secretary, State Pollution Control Board or Committee for the grant of authorization for all or any of the above activities specified in this rule.

(iv) The Member Secretary, State Pollution Control Board or any officer designated by the Board or Committee shall not issue an authorization unless he is satisfied that the operator of a facility or an occupier, as the case may be, possesses appropriate facilities, technical capabilities and equipment to handle hazardous wastes safely.

- (v) The Member Secretary, State Pollution Control Board or any officer designated by the Board or Committee, may, after giving reasonable opportunity of being heard to the applicant refuse to grant any, authorization.
- (vi) The Member Secretary, State Pollution Control Board or any officer designated by the Board shall renew the authorization granted under sub-rule (6),
- (vii) Ever State Pollution control Board or Committee shall maintain a register containing particulars of the conditions imposed under these rules for any disposal of hazardous wastes, on any land or premises and it shall be open for inspection during office hours to any person interested or affected or a person authorized by him in this behalf.

4. Power to suspend or cancel an authorization:

(i) The State Pollution Control Board or Committee may cancel an authorization issued under these rules or suspend it for such period as it thinks fit, if in its opinion, the authorized person has failed to comply with any of the conditions of the authorization or with any provisions of the act or these rules, after giving the authorized person an opportunity to show cause and after

recording reasons therefore.

(ii) Upon suspension or/cancellation of the authorization and during the pendency of an appeal under rule 12, the State Pollution Control Board or Committee may give directions to the persons whose authorization has been suspended or cancelled for the safe storage of the hazardous wastes, and such person shall comply with such directions.

5. Packing, labeling and transport of hazardous wastes:

(i) The occupier or operator of a facility shall ensure that the hazardous wastes are packages, based on the composition in a manner suitable for handling, storage and transport and the labeling and packaging shall be easily visible and be able to withstand

physical conditions and climatic factors.

(ii) Packaging, labeling and transport of hazardous wastes shall be in accordance with the provisions of the rules made by the Central Government under the Motor Vehicles

Act, 1988 and other guidelines issued from time to time.

(iii) All hazardous waste containers shall be provided with a general label as given in Form 8.

(iv) The occupier shall prepare six copies of the manifest in Form 9 comprising of color code indicated below (all six copies to be signed by the transporter):

Copy number with

Purpose

color code

Copy 1 (White)	to be forwarded by the occupier to the State Pollution Control Board or Committee
Copy 2 (Yellow)	to be retained by the occupier after taking signature on it from the transporter and rest of the four copies to be carried by the transporter
Copy 3 (Pink)	to be retained by the operator of the facility after signature
Copy 4 (Orange)	to be returned to the transporter by the operator of facility after accepting waste
Copy 5 (Green)	to be returned by the operator of the facility to State Pollution Control Board/Committee after treatment and disposal of wastes.
Copy 6 (Blue)	to be returned by the operator of the facility to the occupier after treatment and disposal of wastes

(v) The occupier shall forward copy number 1 (white) to the State Pollution Control Board or Committee and in case the hazardous waste is likely to be transported through any transit State, the occupier shall prepare an additional copy each for such State and forward the same to the concerned State Pollution Control Board or Committee before the hands over the hazardous waste to the transporter. No transporter shall accept hazardous wastes from an occupier for transport unless it is accompanied by copy number 2 to 5 of the manifest. The transporter shall return copy number 2 (yellow) of the manifest signed with date to the occupier as token of receipt of the other four copies of the manifest and retain the remaining four copies to be carried and handed over to respective agencies as specified in sub-rule (4).

(vi) In case of transport of hazardous wastes to a facility for treatment, storage and disposal existing in a State other than the State where hazardous wastes are generated, the occupier shall obtain 'No Objection Certificate' from the State Pollution Control Board or Committee of the concerned State or Union Territory Administration where the facility is existing.

(vii) the occupier shall provide the transporter with relevant information in Form 10, regarding the hazardous nature of the wastes and measures to be taken in case of an emergency.

6. Disposal sites:

(i) The occupier or operator of a facility or any association of occupiers shall be jointly and severally responsible for identifying sites for establishing the facility for treatment, storage and disposal of hazardous wastes.

(ii) The State Government, operator of a facility or any association of occupiers shall jointly and

severally be responsible for and identify sites for common facility for treatment, storage and disposal of hazardous wastes in the State.

(iii) The operator of a facility, occupier or any association of occupiers shall undertake an environmental impact assessment (EIA) of the selected site(s) and shall submit the EIA report to the State Pollution Control Board or Committee

(iv) The State Pollution Control Board or Committee shall on being satisfied with the EIA report, cause a public notice for conducting a public hearing as per the procedure contained in the Environmental Impact Assessment Notification, 1994 published vide S.O. 60(E) dated 27th January, 1994 as amended from time to time.

(v) The State Pollution Control Board or Committee shall forward to the State Government or Union territory Administrator, as the case may be, the project report including EIA report and details of public hearing along with its recommendations within a period of 30 days from the last date of public hearing.

(vi) The State Government shall complete the assessment within a period of thirty days from the date of receipt of the documents mentioned in sub-rule (5) and convey the decision of its approval of site(s) or otherwise within 30 days thereafter to the concerned operator of the facility, occupier or any association of occupiers.

(vii) After approval of the site or sites, the State Government shall acquire the site or inform the occupiers to acquire the site(s) for settling up the facility for treatment, storage and disposal of hazardous wastes. The State Government shall simultaneously notify such site(s). The State Government shall also compile and publish periodically an inventory of such hazardous wastes disposal sites and facilities;

(viii) Setting up of an on-site facility for treatment, storage and disposal of hazardous wastes for captive use shall be governed by the authorization procedure laid down in rule 5.

6. Liability of the occupier, transporter and operator of facility:

(i) The occupier, transporter and operator of a facility shall be liable for damages caused to the environment resulting due to improper handling and disposal of hazardous waste listed in Schedule 1, 2 and 3;

(ii) The occupier and operator of a facility shall also be liable to reinstate or restore damaged or destroyed elements of the environment at his cost, failing which the occupier or the operator of a facility, as the case may be, shall be liable to pay the entire cost of remediation or restoration and pay in advance an amount equal to the cost estimated by the State Pollution Control Board or Committee. Thereafter the Board or Committee shall plan can cause to be executed the program for remediation or restoration. The advance paid to State Pollution Control Board or Committee towards the cost of remediation or restoration shall be adjusted once the actual cost of remediation or restoration is finally determined and the remaining amount, if any, shall be recovered from the occupier or the operator of the facility.

(iii) The occupier and operator of a facility shall be liable to pay a fine was levied by the State Pollution Control Board with the approval of the Central Pollution Control Board for any

violation of the provisions under these rules.

7. Appeal:

- (i) An appeal shall lie, against any order of grant or refusal of an authorization by the Member Secretary, State Pollution Control Board or any officer designated by the Board or the Secretary, Department of Environment of the State Government by whatever name called.
- (ii) Every appeal shall be in writing and shall be accompanied by a copy of the order appealed against and shall be presented within thirty days of the receipt of the order passed.
- (iii) Every appeal filed under this rule shall be disposed of within a period of sixty days from the date of such filing.

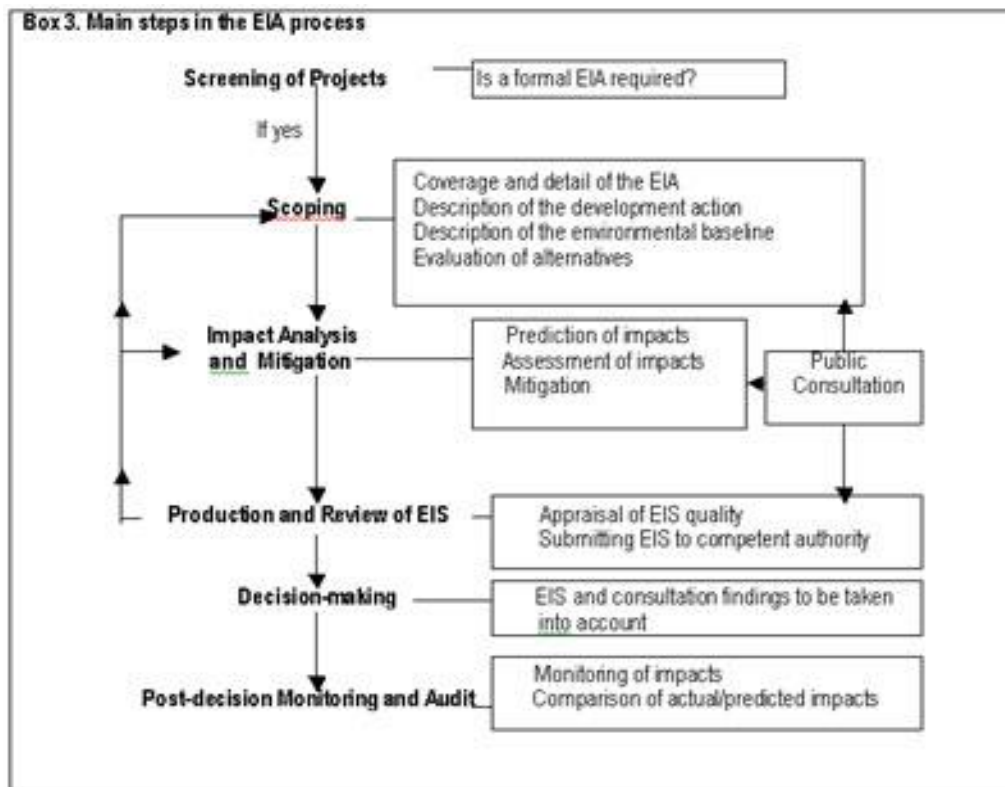
5.2. Environmental Impact Assessment:

5.2.1. Introduction:

An **environmental impact assessment (EIA)** is an assessment of the possible impact—positive or negative—that a proposed project may have on the environment, together consisting of the natural, social and economic aspects.

The purpose of the assessment is to ensure that decision makers consider the ensuring environmental impacts when deciding whether to proceed with a project. The International Association for Impact Assessment (IAIA) defines an environmental impact assessment as "the process of identifying, predicting, evaluating and mitigating the biophysical, social, and other relevant effects of development proposals prior to major decisions being taken and commitments made."

Main steps in the EIA process/Structure of EIA:



Step 1. Screening of Projects

During the screening process the developer has to evaluate whether a formal EIA is required for the project. It is the responsibility of the competent authority to decide whether an EIA is required and then make the decision public. The developer can also decide to voluntarily undertake an EIA without the formal screening decision from the competent authority. Volunteering to undertake an EIA can save time and costs later in the process.

Step 2. Scoping

The scoping stage sets the coverage and detail of the EIA process. Scoping evaluates which impacts and issues to consider and ensures that the impact evaluation provides all the relevant information. Generally scoping takes place between the developer and the competent authority. During the scoping stage those to be consulted, such as communities, local authorities and statutory agencies, are identified. The scoping ought to specify the project in such detail that potential direct, indirect and cumulative impacts can be identified at a later stage.

Description of the project/development action includes information about the purpose and rationale of the project and an understanding of its various characteristics, such as development location and processes.

Description of the environmental baseline should include present and the possible future state of the environment, assuming that the project is not undertaken. The time span should be the same

as for the project. Baseline data project description and mitigation measures should be developed with monitoring implications in mind.

Evaluation of alternatives brings into consideration alternative means of carrying out the project, including technical and technological alternatives. Most of the possible alternatives that arise will be rejected by the developer on economic, technical or regulatory grounds. In the case of unforeseen difficulties during construction or operation of a project re-examination of these alternatives may help to produce rapid and cost-effective solutions.

Step 3. Impact Analysis and Mitigation

During this stage issues identified through scoping are analyzed and the impacts are defined.

The prediction of impacts aims to identify the magnitude and other dimensions of identified change in the environment with or without the project, based on the baseline information gathered during the scoping stage. The significance of impacts could also cast new light on the scoping exercise.

Assessment of impacts assigns relative significance to predict impacts associated with the project, and to determine the order in which impacts are to be avoided, mitigated or compensated.

Mitigation consists of measures to avoid, reduce and if possible to remedy significant environmental effects. At one extreme, prediction and evaluation of impacts can lead to such adverse effects that the only sensible mitigation measure is to abandon the project. Like many elements in the EIA process, mitigation is not limited to one point of assessment.

Step 4. Production and Review of EIS

The environmental information acquired during the assessment is submitted to the Competent Authority by the developer together with the application for development consent. The environmental information is presented in the form of an Environmental Impact Statement (EIS). The EIS is made available to environmental authorities and the public for their information and to obtain their comments. The review involves a systematic appraisal of the quality of the EIS. In some Member States there is a formal requirement for independent review of the adequacy of the environmental information before it is considered by the Competent Authority.

Step 5. Decision-making

The competent authority takes all relevant information (including the EIS and the consultation findings) into account in reaching a decision on the proposed project.

Step 6. Post-decision Monitoring and Audit

Monitoring should include both baseline monitoring (before the project) and impact monitoring (after the project). Post-auditing again involves comparing the impacts predicted in the EIS with those actually occurring after implementation based on the monitoring. This enables an

assessment of the quality of predictions and of the effectiveness of mitigation measures. The main purpose of post-auditing is to provide feedback to the EIA process and apply the lessons learned to be implemented in future EIAs.

Strengths & limitations of EIA

Strengths

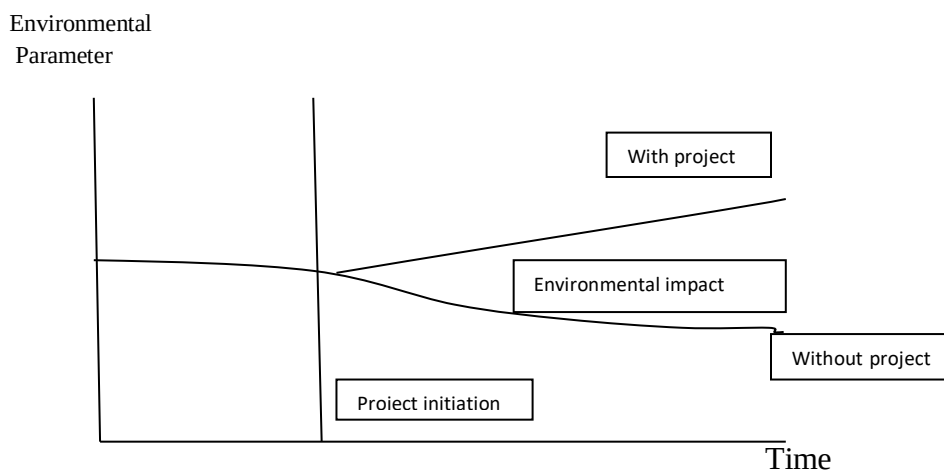
- Improved public participation and co-operation
- Decision-making becomes more transparent
- Universal applicability (many positive outcomes in developing countries)
- Tool for innovations and cost-saving alterations Increases environmental awareness
- Tool for sustainability
- Introduces a cyclical learning process into a linear planning process
- Takes into account trans boundary impacts

Limitations

- Ignores politics and models of decision making
- Uncertainty an intrinsic factor
- The inadequate understanding of the behavior of the environment
- To a great extent a commitment dependent tool
- Susceptible to bias and personal interests (developer as well as pressure groups)
- Quality of data (out of date or the level of detail may be insufficient)

5.2.2. The nature of impacts:

The environmental impacts of a project are those resultant changes in environmental parameters, in space and time, compared with what would have happened had the project not been undertaken. The parameters may be any of the type of environmental receptors noted previously: air quality, water quality, noise levels of local unemployment and crime, for example below figure provides a simple illustration of the concept.



The nature of an environmental impact

The new developments may produce harmful waste streams, but also produce much needed employment in areas of high unemployment. However, the correlation does not always apply. A project may bring physical benefits when, for example, previously polluted and derelict land is brought back into productive use; similarly the socio-economic impacts of a major project on a community could include pressure on local health services and on the local housing market, and increases in community conflict and crime. Projects may also have immediate and direct impacts that give rise to secondary and indirect impacts over time. A reservoir based on a river system not only takes land for the immediate body of water but may also have major downstream implications for flora and fauna, and for human activities such as fishing and sailing.

5.2.3. Classification and identification of impacts:

Impacts are classified into different types, those are

- Positive and negative
- Reversible and irreversible
- Light, moderate and severe

Identification of Impact:

Any economic development project, whether it is a simple and small or a large and complex it has some environmental implications. The environmental implications may be beneficial or adverse, but the main objective of impact identification is to specify areas that are likely to be affected by the implementation of a project. Environmental impact, by definition, implies an alternation of environmental conditions or creation of a new set of adverse or beneficial environmental consequences caused by the action under consideration. Impact identification starts at the early stage of scoping when data on both the project and surrounding environment are made available. As the EIA study progresses, more data become available on the environment and socioeconomic conditions. The preliminary identification of impacts from scoping may be confirmed or new impacts may be identified as requiring investigation.

The identification of new potential impacts may open up new elements of the environment for investigation; the identification of effective measures for mitigating impacts may curtail certain areas of investigation.

5.2.4. Methods of baseline data acquisition:

Baseline information is important reference point for conducting EIA. The term "baseline" refers to the collection of background information on the biophysical, social and economic settings proposed project area. Normally, information is obtained from secondary sources when there is exists a facility of database, or the acquisition of new information through field samplings. The task of collecting baseline data starts right from the period of project inception; however, a majority of this task may be undertaken during scoping.

The form and sources of data vary according to the degree of detail required and the stage in the assessment process. Site-layout diagrams and process – flow charts may be only in outline,

provisional form at the initial design stage. Subsequent investigations, and identification of sources of potential significant impacts, may lead to changes in layout and process. Information describing the site and its environment

Surveys and monitoring are raise a number of issues. They are inevitably constrained by budgets and time, and must be selective. However, such selectivity must ensure that the length of time over which monitoring and survey are undertaken is appropriate to the task in hand. For example, for certain environmental features (e.g. many types of flora and fauna) a survey period of 12 months or more may be needed to take account of seasonal variations. Sampling procedures will often be used for surveys; the extent and implications of the sampling error involved should be clearly established.

The quality and reliability of environmental data vary a great dealt and this can influence the use of such data in the assessment of impacts.

Classification of data:

- “hard” data from reliable sources which can be verified and which are not subject to short-term change, such as geological records and physical surveys of topography and infrastructure;
- “intermediate” data which are reliable but not capable of absolute proof such as water quality, land values, vegetation condition, and traffic counts, which have variable values;
- “soft” data which are a matter of opinion or social values, such as opinion surveys, visual enjoyment of landscape, and numbers of people using amenities, where the responses depend on human attitudes and the climate of public feeling.

A valuable innovation in environmental data provision is the development of computerized data banks, and the increasing use of geographical information systems, for particular sets of data and for particular locations. Such data may, however, have limited accessibility. The analyst should also be wary of the seductive attraction of quantitative data at the expense of qualitative data; both types have a valuable role in establishing baseline conditions. Finally, it should be remembered that all data sources have some uncertainty attached to them, and this needs to be explicitly recognized in the prediction of environmental effects.

Baseline data are collected for two main purposes:

- to provide a description of the status and trends of environmental factors (e.g., air pollutant concentrations) against which predicted changes can be compared and evaluated in terms of importance, and
- to provide a means of detecting actual change by monitoring once a project has been initiated

5.2.5. Impacts on different components:

Assessment of effects:

Effects on human beings, buildings and man-made features:

1. Change in population arising from the development, and consequential environment effects.
2. Visual effects of the development on the surrounding area and landscape.
3. Levels and effects of emissions from the development during normal operation.
4. Levels and effects of noise from the development.
5. Effects of the development on local roads and transport.
6. Effects of the development on buildings, the architectural and historic heritage, archaeological features, and other human artifacts', e.g. through pollutants, visual intrusion, vibration.

Effects on flora, fauna and geology:

7. Loss of, and damage to, habitats and plant and animal species.
8. Loss of, and damage to, geological, paleontological and physiographic features.
9. Other ecological consequences.

Effects on land:

10. Physical effects of the development, e.g. change in local topography, effect of earth-moving on stability, soil erosion etc.
11. Effects of chemical emissions and deposits on soil of site and surrounding land.
12. Land-use/ resource effects:
 - a) Quality and quantity of agricultural land to be taken;
 - b) Sterilization of mineral resources;
 - c) Other alternative uses of the site, including the "do nothing" option;
 - d) Effect on surrounding land uses including agriculture;
 - e) Waste disposal.

Effects on water:

13. Effects of development on drainage pattern in the area.
14. Changes to other hydrographic characteristics, e.g. ground water level, water courses, flow of underground water.
15. Effects on coastal or estuarine hydrology.
16. Effects of pollutants, waste, etc. on water quality.

Effects on air and climate:

17. Level and concentration of chemical emissions and their environmental effects;
18. Particulate matter
19. Offensive odors
20. Any other climatic effects.

Other indirect and secondary effects associated with the project:

21. Effects from traffic (road, rail, air, water) related to the project.

22. Effects arising from the extraction and consumption of materials, water, energy or other resources by the development.
23. Effects of other development associated with the project. Eg. New roads, sewers, housing power lines, pipelines, telecommunications etc.
24. Effects of association of the development with other existing or proposed development.
25. Secondary effects resulting from the interaction of separate direct effects listed above.

5.2.6. Impact assessment methodologies:

The most common formal methods used for impact identification are: checklists, matrices, networks, Overlays and geographic information systems (GIS); expert systems and professional judgment.

Checklists and matrices are commonly used to organize and present information. Many of the more sophisticated methods and techniques often use checklists and matrices as a starting point for analysis.

i) Checklists

Checklists are standard lists of the types of impacts associated with a particular type of project. Checklists methods are primarily for organizing information or ensuring that no potential impact is overlooked. They are a more formalized version of ad hoc approaches in that specific areas of impact are listed and instructions are supplied for impact identification and evaluation.

There are four general types of checklists:

1. **Simple Checklist:** a list of environmental parameters with no guidelines on how they are to be measured and interpreted. These checklists can only help to identify impacts and ensure that impacts are overlooked.

2. **Descriptive Checklist:** includes an identification of environmental parameters and guidelines on how to measure data on particular parameters. They can include data requirements, information sources, and predictive techniques.

Part of a descriptive checklist

Data required	Information sources, predictive techniques
Nuisance: Change in occurrence of odour, smoke, haze etc. and number of people affected.	Expected industrial processes and traffic volumes, citizen surveys.
Water quality: For each body of water, changes in water uses, and number of people affected.	Current water quality, current and expected effluent.
Noise: Change in noise levels, frequency of occurrence, and number of people bothered.	Current noise levels, changes in traffic or other noise sources, changes in noise mitigation measures, noise propagation model, citizen surveys

3. **Questionnaire checklists:** are based on a set of questions to be answered. Some of the questions may concern indirect impacts and possible mitigation measures. They may also provide a scale for classifying estimated impacts from highly adverse to highly beneficial.

Part of questionnaire checklist

Disease vectors:			
a) Are there known disease problems in the project area transmitted through vector species such as mosquitoes, flies, snails, etc.?	Yes	No	Unknown
b) Are these vector species associated with:			
- Aquatic habitats?	Yes	No	Unknown
- Forest habitats?	Yes	No	Unknown
- Agricultural habitats?	Yes	No	Unknown
c) Will the project provide opportunities for vector control			
Through improved standards of living?	Yes	No	Unknown
Estimated impact on disease vectors?			
High adverse	←	insignificant	→ high benefit

4. Threshold of concern checklists: consist of a list of environmental components and, for each component, a threshold at which those assessing a proposal should become concerned with an impact. The implications of alternative proposals can be seen by examining the number of times that an alternative exceeds the threshold of concern.

For example, below table shows part of a checklist developed by the US Forest Service that compares three alternative development proposals on the basis of various components. For the component of economic efficiency, a benefit: cost ratio of 1:1 is the threshold of concern; for spotted owls, 35 pairs are the threshold. In the example, alternative X causes two thresholds of concern to be exceeded, alternative Y one, and alternative Z two; this would indicate that alternative Y is the least detrimental. Impacts are also rated according to their duration: A for 1 year or less, B for 1 – 10 years, C for 10 – 50 years, and D for irreversible impacts. Of the impacts listed, a reduction in the number of spotted owls would be irreversible, and the other impacts would last 10 – 50 years.

Part of a threshold of concern checklist

Environmental component	Criterion	TOC	Alt X		Alt Y		Alt Z	
			Imp	imp>TOC?	Imp	imp>TOC?	Imp	imp>TOC?
Air quality	Emission standards	1	2C	yes	1C	no	2C	yes
Economics	Benefit : cost ratio	1:1	3:1	no	4:1	no	2:1	no
Endangered species	No. of pairs of spotted owls	35	50D	no	35D	no	20D	yes
Water quality	Water quality standards	1	1C	no	2C	yes	2C	yes
Recreation	No. of camping sites	5000	2800C	yes	5000C	no	3500C	yes

There are several major reasons for using checklists:

- they are useful in summarizing information to make it accessible to specialists from other fields, or to decision makers who may have a limited amount of technical knowledge;
- scaling checklists provide a preliminary level of analysis; and
- weighting is a mechanism for incorporating information about ecosystem functions.

ii) Matrices

Matrices require information about both the environmental components and project activities. The cells of the matrix are filled in using subjective (expert) judgment, or by using extensive data bases.

Simple matrices are merely two-dimensional charts showing environmental components on one axis, and development actions on the other. The main advantage of these matrices is the incorporation of cause-effect relationships.

There are, essentially, expansions of checklists that acknowledge the fact that different components of a development project (e.g. construction, operation, decommissioning: buildings, access road) have different impacts. Actions likely to have an impact on an environmental component are identified by placing a cross in the appropriate cell.

Environmental component	Project action				
	Construction		Operation		
	Utilities	Residential and commercial buildings	Residential buildings	Commercial buildings	Parks and open spaces
Soil and geology	X	X			
Flora and fauna	X	X			X
Air quality				X	
Water quality	X	X	X		
Traffic	X	X	X	X	
Employment		X		X	

Three-dimensional matrices have also been developed, in which the third dimension refers to economic and social institutions: such as approach indentifies the institutions from which data are needed for the EIA process, and highlights areas in which knowledge is lacking.

The time-dependent matrix includes a number sequence to represent the time scale (e.g one figure per year) of the impacts. Magnitude is represented by numbers from 0 (none) to 4 (high), over the course of five years.

Environmental component	Project action				
	Construction		Operation		
	Utilities	Residential and commercial buildings	Residential buildings	Commercial buildings	Parks and open spaces
Soil and geology	211	321	0000	0000	0001
Flora	221	422	1223	1111	1123
Fauna	221	311	1100	1100	1122
Air quality	000	000	0123	0034	0011
Water quality	010	022	1223	0111	0000
Traffic	220	332	2333	2333	1111
Employment	120	342	1111	1334	1111

Magnitude matrices go beyond the mere identification of impacts by describing the impacts according to their magnitude, importance, and/or time frame (e.g. short-, medium, or long- term).

Environmental	Project action
---------------	----------------

component	Construction		Operation		
	Utilities	Residential and commercial buildings	Residential buildings	Commercial buildings	Parks and open spaces
Soil and geology	•	•			
Flora	•	●			○
Fauna	•	•			◦
Air quality				•	
Water quality	○	•	•		
Traffic	•	•	•	●	
Employment		○		○	

• = Small negative impact, ● = Large negative impact, ◦ = Small positive impact, ○ = Large positive impact

Leopold Matrix

Leopold et al. (1971) designed a matrix is based on a horizontal list of 100 specified project actions and a vertical list of 88 environmental components. Each action and its potential for impacting each environmental item are considered. The magnitude of the interaction (extensiveness or scale) is described by assigning a value ranging from 1 (for small magnitudes) to 10 (for large magnitudes). The assignment of numerical values is based on an evaluation of available facts and data. Similarly, the scale of importance also ranges from 1 (very low interaction) to 10 (very important interaction). Assignment of numerical values for importance is based on the subjective judgment of the Interdisciplinary team working on the EIA study.

The matrix can also be employed to identify impacts during the various parts of the entire project cycle — construction, operation, and even dismantling phases.

1. Identify all actions (located across the top of the matrix) that are part of the proposed project.
2. Under each of the proposed action, place a slash at the intersection with each item on the matrix if an impact is possible.
3. Having completed the matrix, in the upper left hand corner of each box with a slash, place a number from 1 to 10 which indicates the ‘magnitude’ of the possible impact; 10 represents the greatest magnitude of impact and 1, the least (no zeros). Before each number place + (if the impact would be beneficial). In the lower right hand corner of the box place a number from 1-10 which indicates the importance of the possible impact (e.g. regional Vs local); 10 represents the greatest importance and 1 the least (no zeros)
4. The text which accompanies the matrix should be a discussion of the significant impacts, those columns and rows with large numbers of boxes marked and individual boxes with large numbers.

Environmental components		Proposed actions						
		a	b	c	d	f	g	h
E A R	a							
	b							
	c							

T	d							
H								

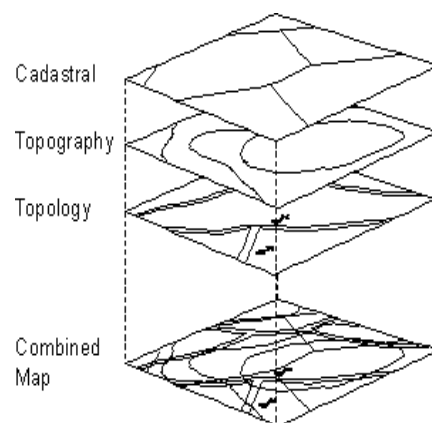
iii) Networks:

Network methods explicitly recognize that environmental systems consist of a complex web of relationships, and try to reproduce that web. Impact identification using networks involves following the effect of development through changes in the environmental parameters in the model. Water is one of the six environmental components, the others being climate, geophysical conditions, biota, access conditions and aesthetics.

iv) Overlay maps:

A series of transparencies is used to identify, predict, assign relative significance to and communicate impacts, normally at a scale larger than local.

Overlay maps are easy to use and understand, and are popular in practice. They are an excellent way of showing the spatial distribution of impacts. They also lead intrinsically to a low – impact decision. The overlay maps method is particularly useful for identifying optimum corridors for developments such as electricity lines and roads, for comparisons between alternatives, and for assessing large regional developments.



5.2.7. Strategies for Risk Assessment:

The development of projects, as e.g. the construction of main roads and railway-lines, the development of industrial plants, shopping centers and theme parks, etc. can cause adverse effects to the environment. EIA and risk assessment are based on very similar concepts and broadly have the same goals. They both deal with the prediction of future consequences arising from activities or planned interventions. They both seek to inform decision-makers about the significance of adverse consequences and are decision-supporting tools that should aid decision-making on measures to mitigate, reduce or eliminate adverse impacts, or the risk of those impacts occurring, respectively.

Environmental risk assessment basically addresses four questions:

- What can go wrong with a project?
- What adverse consequences might occur to human health and the environment?
- What is the range of magnitude of adverse consequences?
- How likely are these consequences?
- What can be done, at what cost, to manage and reduce unacceptable risks and damage?

Methodology: The “triangle-approach”

Consequently, it deals with three different types of data available:

- qualitative data concerning the legal basis and the relevant discussions in the scientific world of EIA policy and application as laid down in the literature;
- quantitative data about the actual application of EIA in the EU Member States deriving from the analysis of the questionnaire; and
- qualitative data about the estimation of the strong and weak points of EIA.

In general, there are three main options available to the risk manager when presented with a risk problem. These options are:

- to reject the intention altogether because it poses unacceptable risks;
- to accept whatever risk is imposed;
- to reduce the risk in some way.

Reducing the risk can be done by doing one or more of the following:

- modifying the hazard;
- modifying the receiving environment;
- modifying exposure;
- modifying the consequences of a risk.

Applications of EIA in terms of risk assessment:

It is useful to identify extraordinary hazards and risks considered in EIA

Which types of extraordinary hazards that could affect a project?

- a. Natural hazards (e.g. flood, earthquake,...)
- b. Accidents in the submitted project due to technological failure (e.g. breakdown)
- c. Accidents in the submitted project due to human failure (e.g. mismanagement)
- d. Accidents in other existing installations in the project's environment

5.2.8. Environmental Impact Statement:

An Environmental Impact Statement is a final analysis or summary of an EIA. “EIS is defined as a documentation of environmental assessment or analysis of a project or action with a potential for environment impacts which are either significant or highly controversial”.

EIS acts as an enforcement mechanism to ensure that the federal government adheres to the goals and policies outlined in the NEPA. An EIS should be created in a timely manner as soon as the agency is planning development or is presented with a proposal for development. The statement should use an interdisciplinary approach so that it accurately assesses both the physical and social impacts of the proposed development.

An EIS typically has four sections:

- An Introduction including a statement of the Purpose and Need of the Proposed Action.
- A description of the Affected Environment.
- A Range of Alternatives to the proposed action. Alternatives are considered the "heart" of the EIS.
- An analysis of the environmental impacts of each of the possible alternatives.

While not required in the EIS, the following subjects may be included as part of the EIS or as separate documents based on agency policy.

- Financial Plan for the proposed action identifying the sources of secured funding for the action. For example, the Federal Highway Administration has started requiring states to include a financial plan showing that funding has been secured for major highway projects before it will approve an EIS and issue a Record of Decision.
- An Environmental Mitigation Plan is often requested by the Environmental Protection Agency (EPA) if substantial environmental impacts are expected from the preferred alternative.
- Additional documentation to comply with state and local environmental policy laws and secure required federal, state, and local permits before the action can proceed.

EIS is a public document written in a format specified by authorized national, state and local agencies. The standard format to be followed in order to prepare an EIS is given below:

1. Project Description

- Purpose of action :
- Description of action : Name & Summary of activities
- Environmental inventory/settling : { Environmental prior to proposed action & Other related governmental activities

2. Land use relationships

- Confirming/confliction with other Land use plans, policies : Country, state and local legislative acts
- Conflicts and or inconsistent land Use plans : { Extent of reconciliation, reasons for proceeding with action

3. Probable impacts of the proposed actions

- Positive or negative impacts : { National or international impacts
Environmental factors
Impact of the proposed action
- Direct and indirect consequences : Primary effects and Secondary effects

4. Alternatives to the proposed action

- Reasonable alternative actions : { Those that might enhance environmental quality
Those that might avoid some/ all adverse effects
- Analyses of alternatives : Benefits, costs, risks.

5. Probable adverse effects that cannot be avoided.

- Adverse and unavoidable impacts

- Avoiding adverse impacts and possible control measures
6. Relationship between local short-term uses of environment and maintenance enhancement of Large- term productivity.
 - Tradeoff between short-term gains at the cost of long-term losses
 - Tradeoff between short-term losses at the cost of long-term gains.
 - Extent to which proposed action controls future options.
 7. Irreversible and irretrievable commitment of resources
 - Unavoidable impacts irreversibly curbing the use of the environment: { Labor, Material
Cultural, Natural
 8. Other interests and considerations of the state policy that offsets the adverse environmental effects of the proposed action.
 - Countervailing benefits of the proposed action & countervailing benefits of the alternatives

Functions of EIS:

The functions that a statement should satisfy are:

- i) It should alert the decision – maker about all aspects of consequences.
- ii) It should help in protecting/ restoring the environment under national policy.
- iii) It is a part of long term planning process.
- iv) It maintains the record of all adverse consequences.
- v) Potential cumulative impacts are realized from EIS.

EIS procedure:

1. Identify the actions: Planning process must lead to firm descriptions of proposed actions before a meaningful assessment is made. Once, the physical activities required to implement a proposed are defined. We must determine whether or not environmental documentation will be required. Therefore, it is first necessary to decide whether the action is major or one that may produce significant effects on the environment.
2. Identify activities involved: Identify detailed activities associated with implementing the project or the program. Those activities not applicable to the project should be rejected and supplementary activities should be added to encompass the project-specific requirement.
3. Examine attributes: The user should examine the attributes and become familiar with the general nature of the individual attributes and the kinds of activities that may on them. In addition attributes may be used to identify areas where available technical expertise is deficient and additional assistance may be required.
4. Evaluate impacts: Using the activities and the attribute list, the user evaluates impacts by constructing a matrix worksheet. Quantify the impact where possible and use an analytic approach for impact analysis.
5. Summarize the impacts: The magnitude of the projects is evaluated along with site characteristics. The degree of severity of impact on each attribute is determined. Finally, the impacts on each attribute from all project activities are summarized.
6. Benefits and costs: The costs are evaluated for the establishment and working of the project. In a similar fashion, the benefits from the proposed project are also evaluated. Benefits here include

both tangible and intangible ones. Based on the benefit-cost ratio the feasibility of the project is analyzed.

7. Review of alternatives: Repeat the above procedure for each of the alternatives considered.

8. Analyze findings: Determine the answers to the following questions concerning the action-

- Will the implementation of the program have a significant adverse effect on the quality of the environment?
- Will the action be deemed environmentally controversial?

If the answer to either of these is 'yes', the proponent is required to prepare an EIS or a "finding of no significant impact (FONSI)" for public records.

9. Prepare assessment document: Depending on whether the decision is to prepare an EIA, or EIS, or a FONSI, the analysis is documented.

10. Process the document: Depending on whether the document is an EIA, an EIS, or a FONSI specific directives for processing or retaining the document are issued by each agency. These directives should be consulted for guidance in processing and coordinating the analysis document. Finally, for the project which requires an EIS, a concise public record for the decision needs to be prepared. This record should include:

- i) What is decision was
- ii) Alternative considered in the discussion
- iii) Techniques to mitigate environmental damage selected for the project.

5.2.9. Concept of Environmental Management Plan:

Environmental management Plan (EMP) is aimed at mitigating the possible adverse impact of a project and ensuring the existing environmental quality. The EMP converse all aspects of planning, construction and operation of the project relevant to environment. It is essential to implement the EMP right from the planning stage continuing throughout the construction and operation stage. Therefore the main purpose of the Environmental Management Plan (EMP) is to identify the project specific activities that would have to be considered for the significant adverse impacts and the mitigation measure required.

EMP During Construction Phase:

The environmental impact during the construction phase will be of short term and reversible nature and will gradually eliminate after the construction activity is over.

EMP During Operation Phase:

During the operation phase, the plant will contribute to environmental pollution in the following manner:

- Atmospheric emission
- Noise Pollution
- Water Pollution and Solid Waste Disposal

EMP Covers the following aspects: i) safeguards and control measures proposed to prevent or mitigate the adverse environmental impacts, ii) plans for rehabilitation of project out sees; iii)

contingency plans for dealing with accidents/ disasters and iv) monitoring and feedback mechanisms on implementation of necessary safeguards.

5.3. Towards Sustainable future:

5.3.1. Concept of Sustainable development:

Sustainable development is defined as development that meets the needs of the present without compromising the ability of the future generations to meet their own needs.

Development is a process that cannot be stopped and should not be stopped.

Sustainable development implies the following factors-

1. Natural resources should be continuously available for man's activities for an indefinite period.
2. Man's scientific and technological achievements should continue to improve the comfort and quality of life.
3. Sustainable development should ensure the preservation of ecological diversity and stability. Man's health and happiness should not be reduced by the developmental process.



Efforts for sustainable development:

To meet the rising needs of the rising population within the limitations imposed by sustainable development, the efforts needed are immense and multi-dimensional.

The primary needs of sustainable development are:

- i. Environmental awareness
- ii. Modified eco- friendly technology
- iii. Determination and will at the individual national and international level and
- iv. Co operation and co ordination.

Prevent global damage is reversible and further damage is preventable, provided the appropriate measures are taken up seriously.

5.3.2. Threats to sustainable development:

- A widening gap between rich and poor, especially the extremes of wealth and poverty
- Poverty: 1.3 billion people live in absolute poverty with incomes of \$1 a day or less, according to the World Bank
- Malnutrition affecting the physical and *mental* development of some 840 million people, according to the UN FAO
- Five hundred companies control 42 per cent of the world's wealth
- The turnover of the world's ten largest companies exceeds the combined GDP of its 100 smallest countries
- Oil company Shell alone controls 160 million hectares of land - 146 countries in the world control a smaller area
- Ten companies effectively control the world's food chain
- Refugees receiving United Nations assistance has exceeded 20 million each year since 1994
- Global warming and climate change - the 1990s is the hottest decade since record keeping began
- The effects of El Nino and La Nina on climate
- Inability of the natural environment to absorb and reprocess mounting quantities of toxic waste
- Pollution which is damaging the habitat on land, at sea and in our backyards; all life depends on habitat quality
- Air traffic is identified as a major cause of greenhouse gases and a contributor to climate change
- Demand for clean, fresh water worldwide is doubling every 21 years
- The hole in the ozone layer, which endangers life, is bigger than ever in 1998
- Over fishing of the oceans which is now threatening the whole web of marine life
- The world's seas are threatened by disputes, destruction, piracy, pollution and sewage
- World grain production struggles to maintain reserves and remains sufficient for less than 60 days use
- Belief that economic recovery can be achieved by increasing demand for goods and scarce resources
- Ageing becomes an increasing problem in many countries as people over 65 represent more than 20% of the population
- Fears that the spread of genetically modified organisms (GMOs) will endanger ecosystems and human health
- Fears that Companies marketing GMOs will patent the seeds and preclude the use of free, non-GMO seeds
- Fears that bio-engineers of big business are seeking control of the evolutionary process
- Failing to realize that companies don't transform, people can and some do.

5.3.3. Strategies or Measures to achieve sustainable development:

1. Population control: Sustainable human population in each area should be estimated and growth should be limited to the desired level.
2. Reduce consumption: Values of life should be shifted to honor lesser consumption and simpler life styles. Mental development should again more respect than material pomp.
3. Integrated land use planning: Relative priorities among different land uses like agriculture, forestry, fodder cultivation, urban and industrial growths, traffic etc. should be planned judiciously.
4. Water resource management: Sustainable development implies optimum management of water resources locally and globally.
5. Renewable resources: Future development should be based on material resources that can be renewed or recycled, or resources cannot be depleted in the near future.
6. Biosphere conservation: Man's activities result in the destruction of species. Thousands of plant and animal species are endangered today with threats of extinction due to pollution, destruction of habitat, commercial exploitation etc. To ensure ecological stability, species diversity is essential.
7. Production efficiency: The current production process involves wastage that can be reduced or avoided. Better planning and improved technology can lead to better yields from given resources.
Genetic hybrids, drip irrigation, pest control, etc. are examples of improving agricultural productivity.
8. Pollution control: Pollution of air, water and soil constitutes a major threat for future mankind. Improved pollution control technology is another major component of sustainable development.

Thus the main measures for sustainable development are:

1. Removing subsidies that encourage excessive use of fossil fuels, pesticides, minerals, timber, water etc.
2. Encourage, educate, empower and involve local people, rural youth and women in their own long term interests.
3. Protect natural habitats and their biodiversity by protecting estuaries, mangroves, wetlands, forest, mountains etc.

5.3.4. Population and its explosion:

The rapid growth of population is the major factor affecting the present and future national and regional development, but it is by no means the only population problem in the world today. The main problem is that of “population explosion”.

The recent rapid increase in human population over the past two centuries has raised concerns that humans are beginning to overpopulate the Earth, and that the planet may not be able to sustain present or larger numbers of inhabitants. The population has been growing continuously since the end of the Black Death, around the year 1400; at the beginning of the 19th century, it had reached roughly 1,000,000,000 (1 billion). Increases in life expectancy and resource availability during the industrial and green revolution led to rapid population growth on a worldwide level. By 1960, the world population had reached 3 billion; it doubled to 6 billion over the next four decades. As of 2009, the estimated annual growth rate was 1.10%, down from a peak of 2.2% in 1963, and the world population stood at roughly 6.7 billion. Current projections show a steady decline in the population growth rate,

with the population expected to reach between 8 and 10.5 billion between the year 2040 and 2050.

The scientific consensus is that the current population expansion and accompanying increase in usage of resources is linked to threats to the ecosystem. The Inter Academy Panel Statement on Population Growth, which was ratified by 58 member national academies in 1994, called the growth in human numbers "unprecedented", and stated that many environmental problems, such as rising levels of atmospheric carbon dioxide, global warming, and pollution, were aggravated by the population expansion. At the time, the world population stood at 5.5 billion, and optimistic scenarios predicted a peak of 7.8 billion by 2050, a number that current estimates show will be reached around 2030.

Population explosion is a sudden increase in the rate of population growth due to a rapid fall in the mortality rate without a corresponding fall in the birth rate.

According to theory of demographic transition, every country passes through 2 stages of demographic transition.

I st stage, both birth and death rates are high and hence population remains more or less stable. The main features of this stage are backward economy where agriculture is the main occupation, low per capita income, low standard of living, inadequate and unbalanced diet, absence of educational opportunities and presence of social evils like polygamy, early marriages etc.

II nd stage, there is a rapid growth of population with the beginning of process of development, the living standards of people improve, education expands, medical and health facilities improve as a result of which death rate comes down. This is called population explosion.

Population explosion doesn't mean overpopulation or population density. Over population or population density is not the major problem. The problem arises when the economic development fails to maintain pace with population growth. The most important factors regarding population are how fast population is growing, and most important is where it is growing.

Ex. Japan has a high population density but it ranked first on the human development index formulated by U.N.D.P. On the other hand low population density areas of Africa or South America are unable to support the existing population.

The, the size, distribution and structure of the population within a country must be viewed in relation to its natural resources and the techniques of production used by its population.

Effects of population:

1. Unemployment
2. Low living standard of people
3. Hindrance in the process of development of economy
4. Pressure on agriculture land
5. Low per capita income
6. Lack of basic needs like water supply and sanitation, education, health etc.
7. High crime rate
8. Environmental damage
9. Migration to urban area in search of job
10. Energy crisis
11. Overcrowding of cities leads to development of slums.

5.3.5. Crazy Consumerism:

The average person is exposed to hundreds of advertisements each day, and the process begins at an early age. Even if the specific product is not remembered, the overall message is: there is a product to meet your every need, if only you will buy it. As Alan has noted,

"People actually remember few ads. Yet commercials have an effect nonetheless. Even if they fail to sell a particular product, they sell consumerism itself by ceaselessly reiterating the idea that there is a product to solve each of life's problems, indeed that existence would be satisfying and complete if only we bought the right things."

Advertisers thus cultivate needs by hitching their wares to the infinite yearnings of the human soul. As Kanner and Gomes point out.

"Large-scale advertising is one of the main factors in American society that creates and maintains a peculiar form of narcissism ideally suited to consumerism. As such, it creates artificial needs within people that directly conflict with their capacity to form a satisfying and sustainable relationship with the natural world."

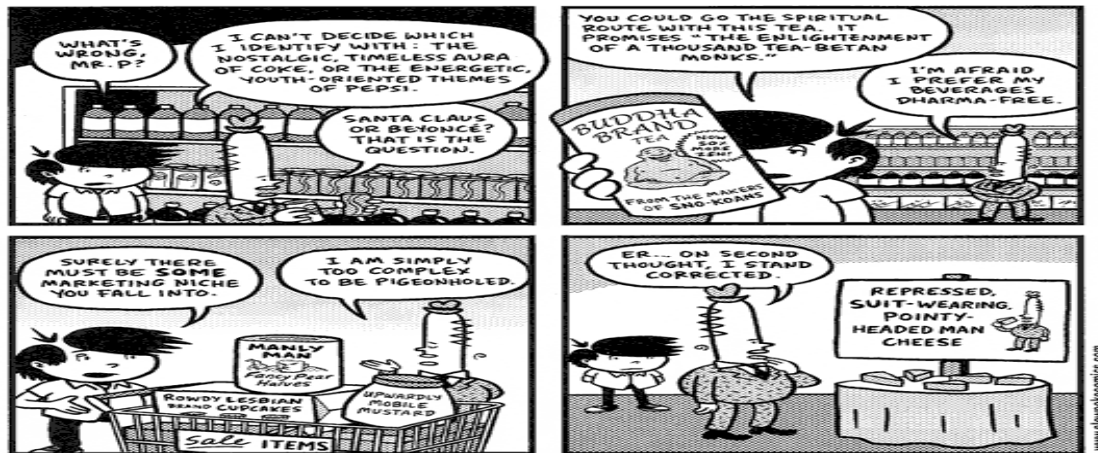
In sum, consumption has become an end in itself, rather than a means to individual enlightenment or happiness, or as a means to social justice, either domestically or globally. And as the consumption habit becomes an end in itself, we have become blind to our own deeper needs.

Human beings are every day and in numerous ways psychologically, socially, and spiritually assaulted by a culture which:

- creates increasing material expectations
- devalues human connectedness
- socializes people to be self-absorbed
- obliterates self-reliance
- alienates people from normal human emotional reactions
- sells false hope that creates more pain

Increasing material expectations: These expectations often go unmet and create pain, which fuels emotional difficulties and destructive behaviors. Many of immigrants found themselves with the pain of increased material expectations that went dissatisfied and they also reported the pain of diminished social support.

Devaluing of human connectedness: Social isolation was highly associated with depression and other emotional problems. Increasing loneliness, however, is good news for a consumer economy that thrives on increasing numbers of "buying units"—more lonely people means selling more televisions, DVDs, psychiatric drugs, etc.



Promotes selfishness. Self-absorption is one of many reasons for U.S. skyrocketing rates of depression and other emotional difficulties—and self-absorption is exactly what a consumer culture demands. The Buddha, 2,500 years ago, recognized the relationship between selfish craving and emotional difficulties, and many observers of human beings

Pain of false hope. The false hope of fundamentalist consumerism is that we will one day discover a product that can predictably manipulate moods without any downsides. Modern psychiatry is a full member of consumer culture. Its "Holy Grail" is a search for the antidepressant that can take away the pain of despair, but not destroy life. In the late 19th century, Freud thought he had found it with cocaine. The antidepressant drug, it is introduced as taking away depression without destroying life. Time after time, it is then discovered that when one tinkers with neurotransmitters, there is—as there is with electroshock and psycho-surgery—damage to life.



5.3.6. Environmental education:

The disciplines included in environmental education are environmental science, environmental engineering and environmental management.

a) Environmental Science: It deals with the scientific study of environmental system (air, water, soil and land), the inherent or introduced changes on organism and the environmental damages incurred as a result of human interaction with the environment.

b) Environmental engineering: It deals with the study of technical processes involved in the protection of the environment from the potentially deleterious effects of human activity and improving the environmental quality for the health and well-being of human.

c) Environmental management: Promotes due regard for physical, social and economic environment of the enterprise or projects. It encourages planned investment at the start of the production chain rather than forced investment in cleaning up at the end. It seeks to develop integrated system rather than unrelated bits and pieces and provides for a good relationship with and concern for the local community rather than paying expensive legal fee to fight it.

Environmental management covers such general areas as environment and enterprise objectives, scope and structure of the environment, interaction of nature, society and the enterprise, environmental impact assessment, economics of pollution, prevention, environmental management standards, environmental audits etc.

Organization of environmental education system:

Environmental education can be imparted in two ways:

I) Formal education

II) Non formal education

D) Formal education: Formal education is given in schools, colleges and universities etc. limited to a specific period, and have a well-defined and systematic curriculum.

The National Council of Educational Research and Training(NCERT) has developed a curriculum frame work based on which many good text books, charts and other teaching aids have designed.

University Grants Commission (UGC) has the main responsibility of environmental education at post graduate level. Environment is a part of the curriculum of mainly the biology courses in Universities and is taught as environmental engineering in many engineering colleges, polytechnic colleges and most Indian Institutes of Technology (IIT's). Environmental Engineering includes subjects like civil engineering, town and country planning, environment improvement of urban slums, human settlements, landscaping, industrial design, design environmental friendly technologies and environmental impact assessment for sustainable development.

Environmental management includes subjects like land-use, agriculture, waste management, wildlife management, conservation of natural resources, forestry, national parks, water resources management, biosphere reserves etc. Besides these main subjects, there are topics dealing with the health and welfare of human beings, for instance, hygiene, toxicology, occupational health, nutrition, chemical engineering etc.

At post graduate level, environmental education also includes social ecology i.e. human ecology, sociology, psychology, environmental ethics etc.

Case studies must be constructed at identified areas as dissertation and field work reports and research must be conducted on specific environmental problems related to the local environment. Environmental legislation is also a part of the curriculum and it includes environmental policies and environmental protection laws etc.

Present situation of formal environmental education in India:

There are nearly two hundred departments of environmental studies in the universities and colleges all over India. They offer degree or diploma programs covering all aspects of

environmental sciences and engineering. There are also diplomas, bachelor's degree, M.Sc, M.Phil and Ph.D programs in environmental sciences or environmental studies. Besides these, M.E., M.Tech and Ph.d programs in environmental engineering are offered by civil engineering and chemical engineering departments.

Post graduate degree programs in environmental management and courses, which lead to M.Sc in environmental chemistry/biology/geology or environmental toxicology, are also available. Environmental education at doctoral level is also available in a large number of autonomous research and development (R & D) institutions founded by central government, state governments and agencies such as CSIR, ICAR and ICMR i.e. formal education in environmental science or engineering is available in India up to the highest possible level.

II) Non- formal education: For a majority of the population that still does not have adequate access formal education, environmental education and awareness can be acquired by programs that fall outside the formal education system.

Environmental education needs to be a lifelong affair rather than a matter of formal schooling. This realization has resulted in an increase in non-formal education which includes activities outside the frame work of the established formal education system. The process of non-formal environment education is experience based involving exercises of solving environment problems. This gives the students an out-of-school exposure which involves the students in natural processes of enquiring, exploring, conjecturing, comparing, inferring, evaluating and decision making regarding environmental problems in their surroundings. Flexibility of approach is the most fundamental characteristic of non-formal environmental education programs.

Non- formal education includes organization of extra-curricular activities like eco development camps, posters and essay-writing competitions, exhibitions, seminars, nature camps, nature-club activities, audio visual slides, mobile exhibitions etc.

Eco – development camps aim at creating awareness about basic ecological principles and solving environmental problems after identifying the causes of the ecological problems. Tree-plantation, trenching, fencing, seed-banks, cleaning water-bodies, hygiene and promoting the use of non- conventional energy sources are the activities included in these camps.

The Vikram Sarabhai Community Centre, Ahamadabad, involved children in carrying out experiments and surveys, and the center for environment education has evolved a water monitoring kit for secondary school students under the Ganga Pollution Control Program.

Arts and Crafts, folk dances, ballet and street plays are also used to impart informal environmental education by many organizations such as Bal Bhawan Society, Shantiniketan.

5.3.7. Urban Sprawl:

Urban sprawl, also known as suburban sprawl, is a multifaceted concept, which includes the spreading outwards of a city and its suburbs to its outskirts to low-density and auto oriented development on rural land, high segregation of uses (e.g. stores and residential), and various design features that encourage car dependency. As a result, some critics argue that sprawl has certain disadvantages, including:

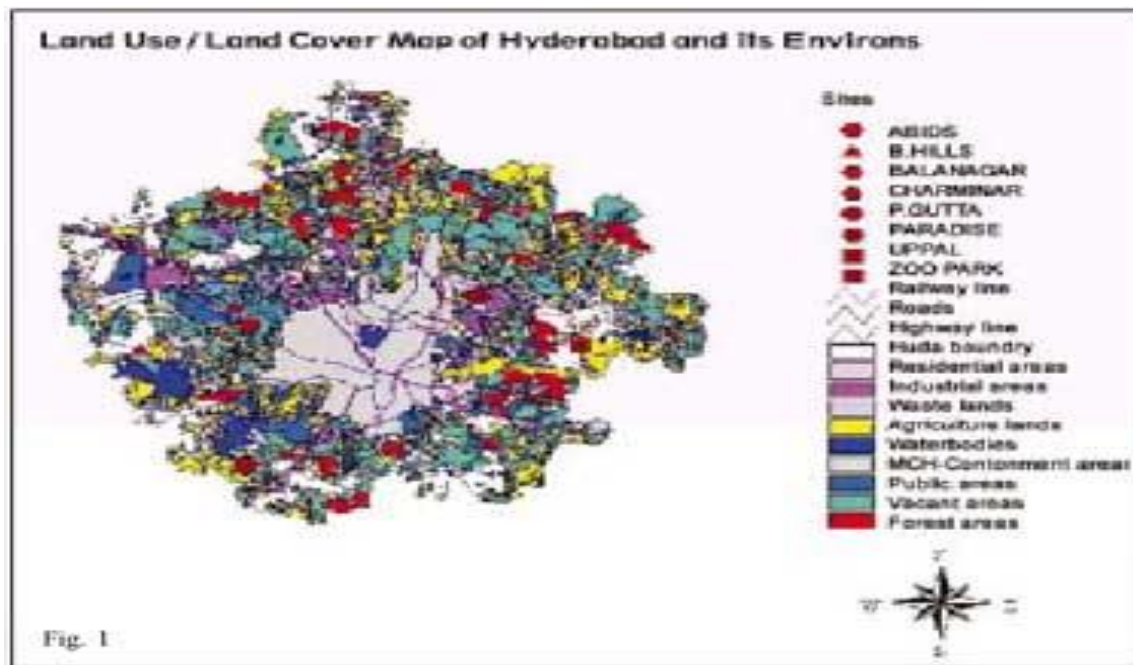
- High car dependence.
- Inadequate facilities e.g.: cultural, emergency, health, etc.
- Higher per-person infrastructure costs.

- Inefficient street layouts.
- Low diversity of housing and business types.
- Higher per-capita use of energy, land, and water.
- Perceived low aesthetic value.

However, critics of the current mainstream of urban planning thought (which has become overwhelmingly anti-"sprawl") respond that sprawl also has certain advantages including:

- More single family residences on larger lots.
- Lower land prices.
- Less experience of noise and pollution.
- Suburban areas generally associated with "sprawl" tend to have lower crime and higher-quality schools.
- Perceived overwhelming consumer preference for sprawl-type developments.

Measuring urban sprawl: A case study of Hyderabad: Remote sensing data is capable of detecting and measuring a variety of elements relating to the morphology of cities, such as the amount, shape, density, textural form and spread of urban areas. Hyderabad occupies fifth position in terms of area and population in the country. The city has been witnessing rapid growth in urban population between 1981 and 1999. The urban population of the city has increased by 41.57% as against 43% of the total Andhra Pradesh state and 36% of total country. In such a scenario, studies on land use cover dynamics over the Hyderabad and its environs gain importance. Land use / cover analysis from the Remote sensing data suggested different land use cover classes viz., residential, industrial, public, semi-public, water bodies and forest (Fig. 1). The detailed land use cover estimates obtained from IRS-1C (LISS III + PAN) data for the years, 1980, 1992 and 1999. Analysis of the results suggests a clear increase in residential, commercial, industrial and transportation in the urban area. In the non-urban area, there is a clear reduction in agriculture area and also in vacant land suggesting the increased intensity with urbanization activities.



5.3.8. Sustainable Cities and Sustainable Communities:

A **sustainable city**, or **eco-city** is a city designed with consideration of environmental impact, inhabited by people dedicated to minimization of required inputs of energy, water and food, and waste output of heat, air pollution - CO₂, methane, and water pollution.

A sustainable city can feed itself with minimal reliance on the surrounding countryside, and power itself with renewable sources of energy. The crux of this is to create the smallest possible ecological footprint, and to produce the lowest quantity of pollution possible, to efficiently use land; compost used materials, recycle it or convert waste-to-energy, and thus the city's overall contribution to climate change will be minimal, if such practices are adhered to.

It is estimated that around 50% of the world's population now lives in cities and urban areas. These large communities provide both challenges and opportunities for environmentally conscious developers. In order to make them more sustainable, building design and practice, as well as perception and lifestyle must adopt sustainability thinking.



Sustainable communities are communities planned, built, or modified to promote sustainable living. They tend to focus on environmental sustainability (including development and agriculture) and economic sustainability. Sustainable communities can focus on sustainable urban infrastructure and/or sustainable municipal infrastructure.

"They [sustainable communities] share a common purpose: places where people thrive to enjoy good health and create a high quality of life. A sustainable community reflects the interdependence of economic, environmental, and social issues by acknowledging that regions, cities, towns and rural lands must continue into the future without diminishing the land, water, air, natural and cultural resources that support them. Housing, transportation and resource conservation are managed in ways that retain the economic, ecological and scenic values of the environment. And they are communities where the consumption of fossil fuels, emissions of greenhouse gases, water resources and pollution are minimized."



5.3.9. Human Health:

Health is the general condition of a person in all aspects... It is also a level of functional and/or metabolic efficiency of an organism, often implicitly human.

At the time of the creation of the World Health Organization (WHO), in 1948, **health** was defined as "a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity".

A major environmental factor affecting health is water quality, especially for the health of infants and children in developing countries.

Maintaining health:

Achieving and maintaining health is an ongoing process. Effective strategies for staying healthy and improving one's health include the following elements:

1. **Observation of Daily Living:** Personal health depends partially on one's active, passive, and assisted observations about their health in their everyday life. The information gleaned from such observations may be used to inform personal decisions and actions (*e.g.*, "I feel tired in the morning so I am going to try sleeping on a different pillow"), as well as clinical decisions and treatment plans (*e.g.*, a patient who notices his or her shoes are tighter than usual may be having exacerbation of left-sided heart failure, and may require diuretic medication to reduce fluid overload) for patients who share their observations with their health care providers.

2. **Social Activity:** Personal health depends partially on the social structure of one's life. The maintenance of strong social relationships is linked to good health conditions, longevity, productivity, and a positive attitude. This is because positive social interaction as viewed by the participant increases many chemical levels in the brain which are linked to personality and intelligence traits.

Volunteering also can lead to a healthy life. To be a volunteer, while gaining plenty of social benefits, people also take their mind off their own troubles. Volunteering could even add years of life. According to a university study, compared with people who did not volunteer, senior citizens who volunteered showed a 67% reduced risk of dying during a seven-year period.

3. Hygiene: Hygiene is the practice of keeping the body clean to prevent infection and illness, and the avoidance of contact with infectious agents. Hygiene practices include bathing, brushing, and flossing, washing hands especially before eating, washing food before it is eaten, cleaning food preparation utensils and surfaces before and after preparing meals, and many others. This may help prevent infection and illness. By cleaning the body, dead skin cells are washed away with the germs, reducing their chance of entering the body.

4. Stress management: Prolonged psychological stress may negatively impact health, and has been cited as a factor in cognitive impairment with aging, depressive illness, and expression of disease. Stress management is the application of methods to either reduce stress or increase tolerance to stress. Relaxation techniques are physical methods used to relieve stress. Psychological methods include cognitive therapy, meditation, and positive thinking which work by reducing response to stress. Improving relevant skills and abilities builds confidence, which also reduces the stress reaction to situations where those skills are applicable.

Reducing uncertainty, by increasing knowledge and experience related to stress-causing situations, has the same effect. Learning to cope with problems better, such as improving problem solving and time management skills, may also reduce stressful reaction to problems. Repeatedly facing an object of one's fears may also desensitize the fight-or-flight response with respect to that stimulus—e.g., facing bullies may reduce fear of bullies.

5. Health care: Health care is the prevention, treatment, and management of illness and the preservation of mental and physical well-being through the services offered by the medical, nursing, and allied health professions.

Workplace wellness programs are recognized by an increasingly large number of companies for their value in improving the health and well-being of their employees, and for increasing morale, loyalty, and productivity. Workplace wellness programs can include things like onsite fitness centers, health presentations, wellness newsletters, access to health coaching, tobacco cessation programs and training related to nutrition, weight and stress management. Other programs may include health risk assessments, health screenings and body mass index monitoring.

Wellness programs may also be found in such places neighborhood community centers and schools. These typically require participants to have a greater degree of commitment to them, as they are voluntary.

6. Public health: Public health is "the science and art of preventing disease, prolonging life and promoting health through the organized efforts and informed choices of society, organizations, public and private, communities and individuals." (Winslow, 1920) It is concerned with threats to the overall health of a community based on population health analysis.

It is well known that the earth's surface and the environment surrounding it are important to human health. The nature of the soil, air, water, temperature, wind, sunshine etc. are determines the man's health and welfare.

The modernization and industrialization has not only uprooted man but in fact has destroyed his habitat and environment too.

Destruction of more forests for extension of land for agricultural purpose, for coal mining activities as well as for making multistoried buildings, roads and other constructions has led to the extinction of some valuable plant and animal species and also responsible for ecological imbalance.

The main diseases suffered by the rural population can be classified as follows:

1. Deficiency diseases: Anemia, Teeth and gum diseases.
2. Arthropodal and helminthic diseases: Malaria, Filariasis, Helmenthic diseases.
3. Communicable diseases: Dysentery and diarrhea, Tuberculosis, Measles etc.
4. Non- communicable diseases: Cancer, Cardiovascular diseases, Lepracy.

5.3.10. Environmental Ethics:

Ethics means the rules of conduct in a system of moral principles. It is a branch of philosophy. As such, every science or profession has its own ethics.

For example- environmental ethics forbid persons from activities that damage his environment like felling of trees.

For obvious reasons, persons who have love, respect and reverence for their environment only can have environmental ethics.

Environmental ethics tell us that we should leave as in-habitable world for our future generations.

Some of the suggestions:

1. Exploitation should stop.
Developed nations with one fourth of the global populations are consuming three fourths of the resources exploited from nature.
2. Emphasis on recycling of materials and renewable and pollution free power resources, should be the key note for the future technology.
3. To promote harmony and give cheer and beauty to human life, the missing components in human growth, namely the aesthetic and spiritual aspects should be restored.
4. Environmental impacts of proposed activities by man, should be assessed by the right expertise.

Environmental ethics are as follows:

- You should love and honor the earth since it has blessed you with life and governs your survival.
- You should keep each day sacred to earth and celebrate the turning of its seasons.
- You should not hold yourself above other living things and have no right to drive them to extinction.
- You should be grateful to the plants and animals which nourish you by giving you food.
- You should limit your offspring's because too many people will overburden the earth.
- You should not waste your resources on destructive weapons.
- You should not run after gains at the cost of nature, rather should strive to restore its damaged majesty.
- You should not conceal from others the affects you have caused by your actions on earth.
- You should not steal from future generations their right to live in a clean and safe planet by impoverishing or polluting it.
- You should consume the material goods in moderate amounts so that all may share the earths precious treasure of resources.

World environment day is celebrated on 5th June.

The concept of 'ahimsa' (non-violence) in Buddhism and Jainism ensure the protection and conservation of all forms of life, thereby keeping the ecological balance of the earth intact.

5.3.11. Concept of green building:

Green building (also known as **green construction** or **sustainable building**) is the practice of creating structures and using processes that are environmentally responsible and resource-efficient throughout a building's life-cycle: from siting to design, construction, operation, maintenance, renovation, and deconstruction. This practice expands and complements the classical building design concerns of economy, utility, durability, and comfort.

Although new technologies are constantly being developed to complement current practices in creating greener structures, the common objective is that green buildings are designed to reduce the overall impact of the built environment on human health and the natural environment by:

- Efficiently using energy, water, and other resources
- Protecting occupant health and improving employee productivity
- Reducing waste, pollution and environmental degradation.

Green building practices aim to reduce the environmental impact of new buildings. Buildings account for a large amount of land use, energy and water consumption, and air and atmosphere alteration. Considering the statistics, reducing the amount of natural resources buildings consume and the amount of pollution given off is seen as crucial for future sustainability, according to EPA.

Green building brings together a vast array of practices and techniques to reduce and ultimately eliminate the impacts of new buildings on the environment and human health. It often emphasizes taking advantage of renewable resources, e.g., using sunlight through passive solar, active solar and photovoltaic techniques and using plants and trees through green roofs, rain gardens, and for reduction of rainwater run-off. Many other techniques, such as using packed gravel or permeable concrete instead of conventional concrete or asphalt to enhance replenishment of ground water, are used as well.

While the practices, or technologies, employed in green building are constantly evolving and may differ from region to region, there are fundamental principles that persist from which the method is derived: Siting and structure Design Efficiency, Energy efficiency, water efficiency Material efficiency, Indoor environmental quality enhancement, Operations and maintenance optimization, and waste and toxics reduction.. The essence of green building is an optimization of one or more of these principles. Also, with the proper synergistic design, individual green building technologies may work together to produce a greater cumulative effect.

On the aesthetic side of green architecture or sustainable design is the philosophy of designing a building that is in harmony with the natural features and resources surrounding the site. There are several key steps in designing sustainable buildings: specify 'green' building materials from local sources, reduce loads, optimize systems, and generate on-site renewable energy.



5.3.12. Ecological Foot Print:

The **ecological footprint** is a measure of human demand on the Earth's ecosystems. It compares human demand with planet Earth's ecological capacity to regenerate. It represents the amount of biologically productive land and sea area needed to regenerate the resources a human population consumes and to absorb and render harmless the corresponding waste. Using this assessment, it is possible to estimate how much of the Earth (or how many planet Earths) it would take to support humanity if everybody lived a given lifestyle. Ecological foot print The Ecological Footprint is used widely as a management and communication tool by governments, businesses, educational institutions, and non-governmental organizations.

What does the Ecological Footprint measure?

The Ecological Footprint measures the amount of biologically productive land and water area an individual, a city, a country, a region, or all of humanity uses to produce the resources it consumes and to absorb the waste it generates with today's technology and resource management practices. This demand on the biosphere can be compared to bio capacity, a measure of the amount of biologically productive land and water available for human use. Biologically productive land includes areas such as cropland, forest, and fishing grounds, and excludes deserts, glaciers, and the open ocean. Ecological Footprint accounts can help us make decisions towards sustainability, and can quantitatively show the positive impacts of groups, businesses, and people making decisions that are helping to bring human demand within the means of the planet.

How is an Ecological Footprint calculated?

Ecological Footprints can be calculated for individual people, groups of people (such as a nation), and activities (such as manufacturing a product). The Ecological Footprint of a person is calculated by considering all of the biological materials consumed, and all of the biological wastes generated, by that person in a given year. These materials and wastes each demand ecologically productive areas, such as cropland to grow potatoes, or forest to sequester fossil carbon dioxide emissions. All of these materials and wastes are then individually translated into an equivalent number of global hectares.

5.3.13. Life Cycle Assessment (LCA):

Life-cycle assessment (LCA, also known as life-cycle analysis, eco balance, and cradle-to-grave analysis) is a technique to assess environmental impacts associated with all the stages of a product's life from cradle-to-grave (i.e., from raw material extraction through materials processing, manufacture, distribution, use, repair and maintenance, and disposal or recycling).

LCAs can help avoid a narrow outlook on environmental concerns by:

Compiling an inventory of relevant energy and material inputs and environmental releases;
Evaluating the potential impacts associated with identified inputs and releases; Interpreting the results to help make a more informed decision

The goal of LCA is to compare the full range of environmental effects assignable to products and services by quantifying all inputs and outputs of material flows and assessing how these material flows impact the environment. This information is used to improve processes, support policy and provide a sound basis for informed decisions.

Life Cycle Inventory (LCI) analysis involves creating an inventory of flows from and to nature for a product system. Inventory flows include inputs of water, energy, and raw materials, and releases to air, land, and water. To develop the inventory, a flow model of the technical system is constructed using data on inputs and outputs.

Inventory analysis is followed by impact assessment. This phase of LCA is aimed at evaluating the significance of potential environmental impacts based on the LCI flow results.

Life Cycle Interpretation is a systematic technique to identify, quantify, check, and evaluate information from the results of the life cycle inventory and/or the life cycle impact assessment. The results from the inventory analysis and impact assessment are summarized during the interpretation phase. The outcome of the interpretation phase is a set of conclusions and recommendations for the study.

LCA also has major roles in environmental impact assessment, integrated waste management and pollution studies.

5.3.14. Low Carbon Life Style:

First of all, we should understand low-carbon. Low-carbon refers to a minimal output of greenhouse gas emissions into the biosphere, specifically refers to the greenhouse gas carbon dioxide. So, low-carbon life means a kind of lifestyle in which people do their best to reduce energy consumption and carbon greenhouse gas emissions. Low-carbon economy is an economy model based on low energy consumption, low pollution and low emission.

Why do we advocate and pursue low-carbon life?

Because of the development of the world industrial economy, the fast increase of population, the unlimited increase of the human desire and the unrestrained production and lifestyle, the human living environment, health and safety are becoming more and more worse, for example, the resources that people rely on to survive dwindle, water pollution becomes worse, the Earth's

ozone layer suffers an unprecedented crisis, the glacier is thawing, the catastrophic global climate changes appear repeatedly. So, we can say, the high-consumption and high-pollution economy model and life style is becoming the killer of the earth and human beings. Therefore, low-carbon life is needed by man's own health, harmony between human beings and nature and the development of mankind.

Then, how can we live a low-carbon life?

Firstly, we should recognize that low-carbon life is not only a kind of lifestyle but also life attitude. Each of us is responsible for the greenhouse gases we emit in our daily actions. Combating climate change has to take the combined efforts of everyone on the planet. Secondly, we should develop good habits in our daily lives. We can select products with longer shelf life when we shop. This can reduce waste and ultimately reduce greenhouse gas emissions. Paper cups are made from cutting forests. We should try not to use them, or select the smaller ones when we do because trees can absorb carbon dioxide and reduce greenhouse gas. We should use efficient light bulbs which use about one-third of the energy of normal bulbs and last 10 times as long. And the lights should be turned off when not needed. This will greatly reduce electricity consumption. We print things out only when necessary, and print double-sided and reduce the margins. All motorized travel is carbon-intensive. Buses and trains are better than cars, but walking or cycling is better still. It's also better for our health and cheaper. Air travel is an environmental disaster. As well as carbon dioxide, planes emit nitrogen oxide, another nasty gas. There are so many low-carbon life actions that it is impossible to list all of them. Low-carbon life can be summarized as moderate food, house, transportation, more sports and not wasting.

5.3.15 Clean development mechanism:

The **Clean Development Mechanism** (CDM) is one of the “flexibility” mechanisms defined in the Kyoto Protocol. The flexibility mechanisms are designed to allow Annex B countries to meet their emission reduction commitments with reduced impact on their economies. The flexibility mechanisms were introduced to the Kyoto Protocol by the US government. Developing countries were highly skeptical and fiercely opposed to the flexibility mechanisms. However, in the international negotiations over the follow-up to the Kyoto Protocol, it has been agreed that the mechanisms will continue.

The purpose of the CDM is to promote clean development in developing countries, i.e., the "non-Annex I" countries (countries that aren't listed in Annex I of the Framework Convention). The CDM is one of the Protocol's "project-based" mechanisms; in that the CDM is designed to promote projects that reduce emissions. The CDM is based on the idea of emission reduction "production". These reductions are "produced" and then subtracted against a hypothetical "baseline" of emissions. The emissions baselines are the emissions that are predicted to occur in the absence of a particular CDM project. CDM projects are "credited" against this baseline, in the sense that developing countries gain credit for producing these emission cuts.

The economic basis for including developing countries in efforts to reduce emissions is that emission cuts are thought to be less expensive in developing countries than developed countries. For example, in developing countries, environmental regulation is generally weaker than it is in developed countries. Thus, it is widely thought that there is greater potential for developing countries to reduce their emissions than developed countries.

From the viewpoint of bringing about a global reduction in emissions, emissions from developing countries are projected to increase substantially over this century. Infrastructure decisions made in developing countries could therefore have a very large influence on future efforts to limit total global emissions. The CDM is designed to start off developing countries on a path towards less pollution, with industrialized (Annex B) countries paying for these reductions.

There were two main concerns about the CDM. One was over the additionally of emission reductions produced by the CDM. The other was whether it would allow rich, northern countries, and in particular, companies, to impose projects that were contrary to the development interests of host countries. To alleviate this concern, the CDM requires host countries to confirm that CDM projects contribute to their own sustainable development. International rules also prohibit credits for some kind of activities, notably from nuclear power and avoided deforestation.

To prevent industrialized countries from making unlimited use of CDM, the framework has a provision that use of CDM be 'supplemental' to domestic actions to reduce emissions. This wording has led to a wide range of interpretations - the Netherlands for example aims to achieve half of its required emission reductions (from a BAU baseline) by CDM. It treats Dutch companies' purchases of European Emissions Trading Scheme allowances from companies in other countries as part of its domestic actions.

The CDM gained momentum in 2005 after the entry into force of the Kyoto Protocol. Before the Protocol entered into force, investors considered this a key risk factor. The initial years of operation yielded fewer CDM credits than supporters had hoped for, as Parties did not provide sufficient funding to the EB. This left it understaffed.

The Adaptation Fund was established to finance concrete adaptation projects and programs in developing countries that are Parties to the Kyoto Protocol. The Fund is to be financed with a share of proceeds from clean development mechanism (CDM) project activities and receive funds from other sources.

Review Questions:

1. What are the basic features of the Water (prevention and control of pollution) Act, 1974?
2. Discuss the basic features of the Air (prevention and control of pollution) Act, 1981.
3. Discuss the role played by the pollution control boards and governments in controlling Water / air pollution.
4. Justify the need of legislation for environmental protection.
5. Name the laws that have been formed for environmental protection and mention the objectives for each Act.
6. What are the penalties for violating the provisions in the Water Act, 1974?
7. Explain in detail about Solid waste management and handling rules.
8. Explain in detail about Hazardous waste management and handling rules.
9. Explain in detail about Bio-medical waste management and handling rules.
10. Explain in detail about Environmental (protection) Act, 1986.
11. Explain in detail about Forest Conservation Act.
12. What is the role of Wild life protection Act in conservation of wildlife.
13. What is environmental impact assessment? Discuss its importance in relation with developmental plans of the country.

14. What are the main impacts of development on environment? How the environmental status may be evaluated?
15. Describe the objectives of environmental management?
16. What are the advantages and disadvantages of EIA?
17. With the help of a flow chart explain the EIA procedure.
18. Define environmental impact statement. How is EIS prepared? Explain.
19. List the different methods adopted for impact assessment. Explain each one of them in detail.
20. Write in detail about green building development.
21. Write a short notes on
A) Classification of Impacts. B) Baseline data acquisition.
22. Explain about Environmental Management Plan.
23. Write a detailed note on sustainable development.
24. Write a short notes on a) Population explosion b) Crazy consumerism
25. Write a note on environmental education in India.
26. Write a short notes on a) sustainable cities and communities b) Urban sprawl.
27. Write a short notes on a) Environmental ethics b) Crazy consumerism.
28. Explain in detail 'Clean Development Mechanism'.
29. Write a short note on a) Ecological Foot Print b) Life Cycle Assessment c) Low Carbon Life Style.
30. Explain about human health.

Multiple choice questions

1. How much water does the average person in the UK use per day? [c]
a) 50 liters (10 buckets) b) 200 liters (40 buckets) c) 150 liters (30 buckets)
2. How much of the UK's tap water is used to flush the toilets daily? [a]
a) 30% b) 20% c) 10%
3. In dry months, how much tap water is used to water the garden in the UK? [b]
a) 10% b) 50% c) 25%
4. How much water do you use in a kettle? [b]
a) Fill it up to the maximum b) Fill it up to the maximum
c) Always boil a little more than actually required, to stop the kettle going dry.
5. How much water do you use for brushing your teeth? [c]
a) Leave the tap running b) Fill the basin up c) A glass or mug is sufficient
6. The roof of an average four bedroom house could catch enough rainwater in a year to [c]
a) Water the garden all year b) Flush the toilets all year c) Both these and much more
7. Rainwater harvesting in your home could save you up to [c]
a) 10% of your mains water b) 20% of your mains water c) 50% of your mains water

8. The Commission on EIA produces advice guidelines. Why are they very often just copied by the competent authority? [c]
- a. since the competent authority does not have the expertise to draw up their own guidelines in many cases
 - b. since civil servants are lazy by definition
 - c. since a competent authority who does not take over the advice guidelines has something to explain
 - d. since the Commission on EIA might decide to ask for supplementary guidelines if their guidelines are not taken over
9. Hydrological models are more frequently applied than ecological models in EIA [a]
- a. True b. not true
10. Which criterion may be used by a Commission for EIA to reject an EIS [c]
- a. The use of expert judgment in the EIS
 - b. The environmental effects are very severe and partly unknown
 - c. The environmental effects are the same for all alternatives
11. SEA is not a separate thing, it is ... [d]
- a. always subject of EIA b. an element in environmental legislation
 - c. always embedded in planning d. embedded in planning or integrated in planning
12. India has a comparative cost advantage in [a]
- a. textiles. b. steel. c. both of them d. none of them
13. Which of the following characteristics are most likely found in developing countries? [d]
- a. high population growth rates. b. large number of people living in poverty.
 - c. very traditional methods of agricultural production. d. all of the above
14. Which of the following could not be considered a major economic system? [d]
- a. capitalism. b. communism. c. socialism. d. physical quality of life index.
15. The Physical Quality of Life Index (PQLI) combines three indicators. They are [a]
- a. infant mortality, life expectancy and adult literacy rate.
 - b. crime rate, clean environment and quality of housing.
 - c. air pollution rate, water pollution rate and sanitation.
 - d. health, education and environment.
16. Which of the following applies to 'externalities'? [c]
- a. They are the private costs of economic behavior b. they are always negative.
 - c. They are not normally reflected in the market price of a product.
 - d. they are always responsible for a welfare loss.
17. Which of the following statements is correct? [b]
- a. Sustainable development is another name for economic growth.
 - b. Sustainable development is not necessarily incompatible with economic growth.
 - c. Sustainable development implies that economic growth will have to become negative.
 - d. Sustainable development is concerned solely with economic issues.
18. The disease that is non-communicable is _____. [b]
- a. malaria b. Marasmus c. AIDS d. Jaundice

19. Malaria is caused by a _____. [a]
a. protozoan b. fungi c. virus d. bacteria
20. The husk of the isabgol seed with water produces relief from _____. [d]
a. malaria b. jaundice c. flu d. diarrhea
21. The vitamin that is not fat soluble is _____. [b]
a. vitamin A b. vitamin B complex c. vitamin D d. vitamin E
22. An insect which transmits a disease is known as _____. [c]
a. intermediate host b. parasite c. vector d. prey
23. A chronic case of a disease denotes _____. [c]
a. severe attack of the disease b. mild attack of the disease
c. disease occurs for a very long period d. all of the above
24. Which one is a communicable disease? [a]
a. Malaria b. Diabetes c. Hypertension d. Helminth
25. Which one of the diseases is not communicable? [d]
a. Typhoid b. Leprosy c. Measles d. Leukemia
26. AIDS virus has _____. [c]
a. single strand DNA b. double strand DNA
c. single strand RNA d. double strand RNA
27. Immuno-deficiency syndrome could develop due to _____. [c]
a. defective liver b. defective thymus
c. AIDS virus d. weak immune system
28. T.B. is cured by _____. [c]
a. griseofulvin b. ubiquinone c. streptomycin d. encitol
29. AIDS is due to _____. [a]
a. reduction in number of helper T-cell b. reduction in number of killer T-cell
c. auto-immunity d. non-production of interferon's
30. Typhoid is caused by _____. [c]
a. Escherichia b. giardia c. salmonella d. shigella
31. Which of the following is a mismatch? [b]
a. Leprosy - Bacterial infection b. AIDS - Bacterial infection
c. Malaria - Protozoan infection d. Elephantiasis - Nematode infection
32. Fever, delirium, slow pulse, abdominal tenderness and rose coloured rash indicate the disease _____. [a]
a. typhoid b. measles c. tetanus d. chicken pox
33. Calcium deficiency occurs in the absence of vitamin _____. [a]
a. D b. A c. C d. B
34. . Which of the following is not an example of public policy? [b]

- a) Environmental regulation b) Unifying religious organizations
- c) Achieving societal goals d) Regulating industry

35. Which of the following does not have responsibility for the formulation of public policy? [c]

- a) Federal agencies b) The president c) The public d) Congress

36.. Pollution credits were created by : **the Clean Air Act of 1990.**

37. Which is directly or indirectly related to essentially all of our environmental problems?
population growth

38. Which of the following occurred during the 1970s? [d]

- a) passage of the Clean Air Act b) passage of the Endangered Species Act
- c) passage of the Clean Water Act d) all of the above

39. Which of the following global trends is of great concern for the future of our environment? [d]

- a) species extinction leading to the loss of biodiversity b) degradation of fertile soils
- c) population growth and increasing per capita consumption of resources
- d) all of the above

40. In India, which one of the following Acts was enacted in the year 1995? [c]

- (a) The Environment (Protection) Act (b) The I-breaths (Conservation) Act
- (c) The National Environment Tribunal Act (d) The Wildlife (Protection) Act

41. Geographical Information Science (GISc) can be defined as: **the science behind GIS**

42. Performing the same analysis in two different GIS software packages will always give the same results. True/**False**

43. Reality can be represented in GIS as a series of layers or as objects. **True/ False**

44. Attribute data are one type of spatial data. True/**False**

Model Paper – I

Time: 3 hrs

Max Marks: 75

Note: This question paper contains two parts A and B. Part A is compulsory which carries 25 marks. Answer all questions in part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a,b,c as sub questions.

Part – A

1. a. Define bio-magnification. 2M
- b. Explain food chains with examples. 3M
- c. What are the manmade factors for drought? 2M
- d. What are the problems of mineral exploration? 3M
- e. Explain aesthetic value of biodiversity. 2M
- f. Define biodiversity hot-spots. 3M
- g. Define primary and secondary pollutants. 2M
- h. Write a note on ozone depleting substances. 3M
- i. Define EIA. 2M
- j. What is Crazy consumerism? 3M

Part – B

2. a) Write structure of an ecosystem.
- b) Explain energy flow in the ecosystem.
- c) Give classification of ecosystems.

Or

3. a) Define biogeochemical cycles.
- b) Explain detailed cyclic flow of Nitrogen between biotic and abiotic components with the help of neat diagram.
- c) Write about carrying capacity of an ecosystem.

4. a) Write a note on problems of over exploitation of natural resources.
- b) Describe about advantages and disadvantages of Dams.

Or

5. a) Discuss about alternative sources of energy.
- b) Explain about land degradation.

6. a) Define genetic diversity.
- b) Explain about consumptive use value of biodiversity.

Or

7. a) Write a note on major threats to biodiversity.
- b) Discuss about National Biodiversity Act.

8. a) Give the Ambient air quality Standards.
b) Write about impacts of modern agriculture on soil.
c) Explain about secondary treatment of waste water.

Or

9. a) Write a note on climate change on human environment.
b) What are the problems of Ozone layer depletion?
c) What is the role of Kyoto Protocol in control of Global Warming?

10. a) Write a note on Environmental Protection Act.
b) Define water pollution under Water pollution Control Act.
c) What are the steps involved in EIA process.

Or

11. a) Define EMP.
b) Explain about threats to sustainable development.
c) Write a note on Life Cycle Assessment.

@@@@@@@@@@@

Model Paper - 2

Time: 3 hrs

Max marks: 75

Part- A

(25M)

1. a) Explain about food web. 2M
- b) Define Saprophytes and Detrivores. 3M
- c) Distinguish between living and non-living natural resources. 2M
- d) Define ground subsidence and write its problem. 3M
- e) Define biotic communities' diversity. 2M
- f) Write a note on man and wildlife conflicts. 3M
- g) Define bioremediation. 2M
- h) What are the anthropogenic sources of desertification? 3M
- i) Distinguish between Solid waste and Bio-medical waste. 2M
- j) Define population explosion and write its problem. 3M

Part – B

2. a) Explain about energy flow in the ecosystem.
- b) Explain about Carbon Cycle with the help of a neat diagram.

Or

3. a) Write a note on Ecological Pyramids.
- b) Write a field report on any natural ecosystem,.

4. a) What are the problems of Dams, explain with the help of a case study.
- b) What are the environmental effects of mineral exploration.

Or

5. a) Explain in detail about soil erosion.
- b) Explain about problems of Timber and Wood extraction on Ecological Balance.

6. a) Write a note on biodiversity hot-spots.
- b) Explain about conservation of biodiversity.

Or

- 7.a) Explain about productive use value of biodiversity.
- b) Write a report on Zoo Park Visit.

8. a) What are the sources of air pollution.
b) Explain about depletion of ozone layer by CFC's with a neat diagram.
c) Explain about Montreal Protocol.

Or

9. a) What are the sources and effects of climate change.
b) Write a note on drinking water quality standards.
c) What are the sources and characteristics of e – waste.
10. a) Explain about Forest Conservation Act.
b) What are the functions of pollution control boards under Water pollution control Act.
c) What are the strategies for risk assessment in EIA.

Or

11. a) Write a note on Environmental education system in India.
b) Explain about Carbon Foot Print.
c) Explain about Green Building Development.

@@@@@@@@@